Retrofitting car design of an Internal Combustion engine car into Electric Vehicle

A

Project Work

Submitted in partial fulfillment of the requirement for the award of the degree of

Bachelor of Technology

In

MECHATRONICS ENGINEERING

Submitted by

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Accredited by NBA, New Delhi and NAAC with A++ Grade

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CERTIFICATE

This is to certify that the project entitled **"Retrofitting car design of an Internal Combustion engine car into Electric Vehicle"** that is being submitted by **MOHAMMED MISBAHUDDIN ADNAN, AARON JOSEPH JOJI, MULLA MOHAMMED FAIZ MUBARAK, PEERJADE SAYAD GAFOOR** bearing the Roll number **21261A1426, 22261A1401, 22261A1425, 22261A1427** of "**BACHELOR OF TECHNOLOGY**", Department of Mechanical (Mechatronics) Engineering specialization in "**MECHATRONICS**" is a project work carried out by us during the academic year 2025- 2026.

The result embodied in this project report has not been submitted to any other university or institute for the award of any degree.

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No amount of words can adequately express the unrelenting efforts taken by our parents in bringing me up to acquire a Bachelor's Degree in Engineering and we thank them for bringing me up with good virtues and morals.

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DECLARATION

WE, MOHAMMED MISBAHUDDIN ADNAN, AARON JOSEPH JOJI, MULLA MOHAMMED FAIZ MUBARAK, PEERJADE SAYAD GAFOOR, hereby declare that the project work entitled, "**Retrofitting car design of an Internal Combustion engine car into Electric Vehicle"** is an authenticated work carried out by us, for the partial fulfillment of the award of the degree of Bachelor of technology, at the department of Mechanical Engineering(Mechatronics) **Mahatma Gandhi Institute of technology**, Hyderabad.

The content of this thesis in full or parts have not been submitted to any institute or university for the award of any degree.

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ABSRACT

With fuel prices rising steadily and vehicular pollution becoming a growing concern in urban India, need to transition towards cleaner mobility solutions has never been more urgent. While purchasing a brand-new Electric Vehicle remains financially out of reach for many, retrofitting an existing Internal Combustion Engine (ICE) vehicle into an EV presents a practical and cost-effective alternative that extends the life of older vehicles while significantly reducing their environmental footprint.

This project undertakes the complete retrofit design of a Maruti 800, one of India's most widely available and lightweight small cars, converting it from petrol powered engine into a functional Electric Vehicle. The conversion involves removing the 3-Cylinder petrol engine, fuel system, exhaust, and associated components, and replacing them with 10kW PMSM Motor, a 48V lithium-ion battery pack, a motor controller, a battery management system (BMS), a DC-DC converter for auxiliary system, and a regenerative braking system.

The project covers the complete engineering workflow from theoretical force and power calculation (rolling resistance, aerodynamic drag, and gradient forces) to component sizing, parts selection, and a simulation of the drivetrain. The simulation models vehicle speed, motor torque, G-force response, and motor temperature under real driving conditions, while theoretical and actual calculations and compared side by side to validate the design choices.

The goal is not just to build a converted vehicle, but to establish a documented, repeatable methodology that could serve as a reference for similar low-cost EV retrofit projects across India, contributing toward sustainable urban mobility without the financial burden of new EV adoption.

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CHAPTER 1: INTRODUCTION

The global automotive sector is currently passing through one of the most significant technological transitions since the invention of the internal combustion engine (ICE). For more than a century, ICE vehicles have dominated personal and commercial mobility, but their widespread use has created severe environmental, economic, and sustainability challenges. With rapidly rising fuel prices, increasing geopolitical dependency on petroleum, degraded urban air quality, and accelerating climate change, it has become obvious that traditional fossil-fuel based mobility is no longer viable for the future. Governments worldwide are enforcing stricter emission regulations, industries are pushing for cleaner technologies, and consumers are becoming more environmentally conscious. In this urgent push toward sustainable mobility, electric vehicles (EVs) have emerged as one of the most disruptive and promising alternatives.

However, the transition from ICE vehicles to EVs is not straightforward. New EVs are still relatively expensive for the average buyer in developing nations like India. Their purchase cost, combined with battery replacement expenses, makes adoption slow despite long-term operational savings. At the same time, India has a massive population of ageing ICE vehicles millions of them still structurally usable but environmentally harmful. Replacing all these vehicles with brand-new EVs is neither economically realistic nor environmentally optimal. Manufacturing a new vehicle generates significant carbon emissions; therefore, scrapping a functional vehicle only to buy a new EV contradicts the principles of sustainability and circular economy.

This is where retrofitting or converting existing ICE vehicles into Electric Vehicles becomes a practical, cost-effective, and environmentally powerful solution. Retrofitting extends the life of an existing car, eliminates tailpipe emissions instantly, cuts operational costs drastically, and avoids the massive carbon footprint of manufacturing a new vehicle. For countries with large middle-class populations and strict budgets, such as India, retrofitting is arguably the fastest way to accelerate EV adoption without forcing consumers into expensive purchases.

In this project, the team focuses on studying, analysing, and partially implementing the conversion of a small ICE passenger vehicle specifically a Maruti 800 into a fully electric vehicle. The Maruti 800 is chosen not randomly, but because it is one of the lightest, simplest, most widely available and mechanically straightforward vehicles in India. Its chassis layout, low kerb weight, simple transmission architecture, and large availability in second-hand markets make it an ideal candidate for educational as well as real-world EV conversion.

The project examines the complete technical, economic, and environmental landscape of retrofitting. It studies what parts must be removed engine, fuel tank, exhaust, radiator and what electric components must be added motor, controller, battery pack, auxiliary electronics. It evaluates the forces acting on the vehicle, the power required to propel it at realistic speeds, and the energy demand needed to reach usable driving ranges. Through detailed calculations, MATLAB simulations, and literature surveys, the project determines a suitable motor rating (10kW), required torque (~132 Nm), battery capacity (10.24 kWh), and overall performance metrics after conversion.

The study does not simply model the propulsion requirements but also exposes the challenges associated with retrofits thermal management, regenerative braking tuning, battery packaging, wiring safety, and regulatory limitations in India. It compares retrofitting against buying a new EV, evaluates cost-effectiveness, and identifies technical gaps that prevent retrofits from being mainstream despite their environmental advantages.

At a broader level, this project aligns with India's mission toward sustainable mobility. India is among the fastest-growing automotive markets in the world, yet also one of the most polluted. Vehicle emissions contribute heavily to PM2.5 pollution, urban heat stress, and respiratory disease. Retrofitting old ICE vehicles into EVs can immediately reduce emissions without waiting for new EVs to penetrate the market at scale. It is a direct route to decarbonization, extending vehicle life while eliminating fuel costs.

Therefore, this project is not just an academic study it is a practical exploration into a technology that has the potential to transform India's mobility landscape. By combining engineering calculations, simulation tools like MATLAB, real-world case studies, and feasibility analysis, the project provides a structured, technical foundation for future work on EV retrofits.

1.1 Motivation of the Project

The motivation behind this project arises from the urgent need to reduce pollution and dependency on fossil fuels while keeping solutions affordable and accessible. New EVs are financially out of reach for a majority of the Indian population. However, millions of older vehicles can be converted into clean electric ones at a fraction of the cost. This offers an immediate and scalable pathway to sustainable mobility, without waiting for complete market transformation. Additionally, retrofitting supports circular economy principles, reduces manufacturing waste, and gives ageing vehicles a second life all while cutting down fuel expenses dramatically.

1.2 Problem Definition

The primary problem addressed in this project is the lack of cost-effective, technically sound, and scalable methods to convert existing ICE vehicles into efficient electric vehicles. Key challenges include selection of appropriate motor and battery size, handling vehicle dynamics after conversion, ensuring safety of high-voltage components, optimizing efficiency, and evaluating the feasibility of the conversion both economically and environmentally.
The project aims to systematically analyse these issues through calculations, literature study, and MATLAB-based simulation.

1.3 Scope of the Project

The scope of this work includes:

- Studying the feasibility of converting a small ICE car into an EV.
- Identifying vehicle components to be removed and replaced.
- Determining required motor torque, power, and battery capacity using engineering calculations.
- Simulating the converted vehicle under real driving conditions using MATLAB models.
- Reviewing similar retrofit projects, conversion kits, and Indian case studies.
- Evaluating cost-effectiveness compared to purchasing a new EV.
- Highlighting limitations, challenges, and future opportunities in retrofitting.

CHAPTER 2: LITERATURE REVIEW

The rapid shift toward sustainable mobility has led researchers, engineers, and policymakers across the world to explore alternatives to conventional internal combustion engine (ICE) vehicles. One of the most promising pathways identified in recent years is the conversion, or *retrofitting*, of existing ICE cars into Electric Vehicles (EVs). While new EVs continue to evolve technologically, their high cost, limited availability, and long payback periods prevent immediate large-scale adoption, especially in developing nations like India. Retrofitting emerges as a practical, cost-effective, and environmentally efficient solution that reuses an existing vehicle platform while eliminating tailpipe emissions. The literature on EV retrofits spans multiple domains including drivetrain engineering, battery technology, regulatory frameworks, environmental impact analyses, and real-world case studies which collectively define the feasibility of this approach.

2.1 Global and Indian Context of EV Retrofitting

In advanced markets, EV retrofitting initially gained popularity among classic car enthusiasts, who desired to preserve heritage vehicles while modernizing their propulsion systems. However, in India, the motivation is more pragmatic reducing pollution, lowering fuel costs, and extending the usable life of older vehicles. Studies from the Global Green Growth Institute (GGGI, 2023) emphasize that retrofitting can reduce lifecycle greenhouse gas emissions by 35–60% compared to continuing ICE vehicle usage. These findings align directly with India's sustainability goals and pollution control measures, especially in cities that enforce scrappage policies.

2.2 Real-World EV Conversion Case Studies

One of the most cited Indian case studies is the Team-BHP (2020) documentation of a Maruti 800 converted into a fully functional electric car.
Key technical features include:

- 19 kW BLDC motor
- 13.2 kWh Winston LiFePO4 battery pack
- Curtis programmable controller
- Custom single-speed reduction transmission

- Regenerative braking with hill-hold functionality
- Modified suspension and widened track. The converted vehicle achieved 120 km range and 80–85 km/h top speed, proving that a small ICE vehicle can realistically be converted into a usable EV without factory-level manufacturing tools. This case validates the technical feasibility and serves as a benchmark for academic projects like the present one.

Another example referring from Mechatronic Trading (2023), a manufacturer offering conversion kits for cars such as Maruti 800 and Tata Nano. Their kit includes:

- A 60 V, 3.5 kW BLDC motor
- Throttle pedal, controller, wiring harness
- Motor mounts and differential options

  Although modest in power, the kit demonstrates that standardized retrofit solutions are entering the Indian market, paving the way for scalable conversions.

2.3 Technical Approaches to ICE-to-EV Conversion

Powertrain Replacement:

Retrofitting involves removing the ICE, fuel tank, exhaust, radiator, and transmission components, replacing them with:

- Electric motor (BLDC, PMSM, induction)
- Motor controller/inverter
- Battery pack
- DC-DC converter
- Wiring harness
- Charging port

Taking Eydgahi (2011) as reference, it notes that retaining the existing gearbox simplifies integration, reduces mechanical complications, and gives the converted EV better torque multiplication at low speeds. Modern PMSM motors combined with Field-Oriented Control (FOC) increase efficiency and responsiveness.

Drivetrain-Integration:

Engineering decisions direct drive vs gearbox, motor placement, torque requirements determine the conversion quality. Retrofits commonly use indirect-drive systems due to simplicity and compatibility with existing vehicle architecture.

Performance Considerations:

Conversion performance hinges on:

- Vehicle mass after ICE removal
- Drag coefficient
- Rolling resistance
- Motor torque-speed characteristics
- Battery chemistry and energy density

Typical small-car conversions in literature use motors between 10–30 kW, yielding performance comparable to city-driving ICE cars.

2.4 Environmental and Life-Cycle Considerations

Multiple life-cycle analyses (LCA) show that the environmental burden of producing a new EV especially the battery can be offset through the operational phase. However, retrofitting eliminates the need to manufacture an entirely new vehicle, dramatically reducing:

- Embedded emissions in steel, aluminium, and plastics
- Manufacturing energy consumption
- Logistics and transportation-related emissions

Taking Hoeft (2021) as reference, it notes that a retrofitted EV reduces material demand and supports the circular economy by using existing chassis infrastructure. Additionally, using second-life batteries further lowers cost and environmental impact, though at the expense of reduced range and uncertain lifetime.

2.5 Regulatory and Safety Framework

The U.S. DOE Alternative Fuels Data Center outlines guidelines on electrical safety, battery mounting, crashworthiness, and electromagnetic compatibility (EMC). Countries like the UK and EU have well-defined homologation pathways for EV retrofits.
In India, however, the regulatory landscape is still evolving. Key challenges include:

- Lack of standardized certification for retrofit kits
- Safety validation of high-voltage systems
- Insurance and registration complications
- Structural integrity assessments

Despite these gaps, recent government discussions and EV policy drafts indicate a growing push toward formalizing retrofit regulation.

2.6 Economic Feasibility in Literature

Economic feasibility varies significantly across studies:

- Hobbyist retrofits with premium components are expensive.
- Modular kits with standardized components reduce cost drastically.
- Fleet retrofits (taxis, delivery vans) are found to be most economical due to frequent daily usage.

Taking River Publishers (2023) as reference, it highlights that the main cost drivers are:

- Battery pack (≥40% of total cost)
- Motor and controller
- Fabrication and labor
  The overall cost is still far below purchasing a new EV, making retrofitting the most practical near-term solution for low-income markets.

2.7 Performance, Range, and Component Sizing

A crucial outcome across many studies is that correct sizing of motor and battery determines the success-of-the-retrofit.
Key guidelines from literature include:

- A small city car requires 5–15 kW continuous power.
- Peak power depends on gradient climbing and acceleration requirement.
- Battery capacity between 6–12 kWh offers 50–120 km range.
- Weight distribution significantly affects handling and braking.

Theoretical and actual calculations from this project fall exactly in this range, validating the correctness of your motor power (~7 kW) and battery sizing (~8.17 kWh).

2.8 Technical Challenges Highlighted in Literature

Common challenges include:

- **Thermal management:** Many retrofits lack proper cooling systems.
- **Regenerative braking tuning:** Poor control causes jerky deceleration.
- **Instrumentation:** Basic meters fail to log accurate battery or motor data.
- **Battery packaging:** Older cars lack space designed for prismatic cells.
- **High-voltage safety:** Improper insulation risks short circuits or fire.

2.9 Global Trends in EV Retrofitting

Retrofitting is no longer an experimental concept; it is becoming a mainstream strategy in many countries. In Europe, companies such as Transition-One (France), Lunaz (UK), and Electrogenic (UK) offer standardized retrofit kits for small cars, classic vehicles, and even commercial fleets. These conversions are being supported through government incentives and relaxed homologation rules. The European Union has also published frameworks encouraging the use of refurbished EV components, especially for fleet vehicles.
In the United States, the Department of Energy encourages conversion of light commercial

vehicles and school buses due to their high daily usage and predictable routes. Similar trends are emerging in Palestine, Australia, and Japan.
These international efforts demonstrate that retrofitting is now recognized as a scalable strategy not just for enthusiasts but as a serious component of sustainable mobility.

2.10 Battery Technology Evolution and Its Impact on Retrofitting

Battery advancements significantly influence the feasibility of EV conversions. Modern lithium iron phosphate ($LiFePO_4$) and NMC (Nickel Manganese Cobalt) batteries provide higher energy density, longer lifecycle, and greater safety than older lead-acid or NiMH chemistries. Research on solid-state batteries promises even higher density and lower risk of thermal runaway.
For retrofit applications, battery packaging is one of the greatest challenges: older ICE chassis were not designed to hold large lithium packs. Studies recommend modular battery design, distributed battery placement, and the use of standardized cell formats such as 18650, 21700, or prismatic LFP cells.
Battery management systems (BMS) are also crucial. Literature highlights the importance of cell balancing, thermal protection, current limiting, and state-of-charge estimation for ensuring safe retrofit operations.

2.11 Motor Technology Advancements Relevant to Retrofitting

Electric motor technology has advanced rapidly in the last decade. Permanent Magnet Synchronous Motors (PMSMs) and high-efficiency Brushless DC Motors (BLDCs) now dominate the EV conversion market due to their superior torque density and efficiency. Induction motors, while cheaper, are bulkier and require more complex controllers.
Recent studies compare FOC (Field-Oriented Control) with traditional trapezoidal control, showing that FOC significantly improves torque smoothness and reduces energy losses.
Lightweight, high-torque axial flux motors are emerging as a promising option for compact EV conversions, showing higher power density than traditional radial flux motors.
These innovations make it possible to retrofit even small cars like Maruti 800 without compromising drivability.

2.12 Retrofitting Challenges Highlighted in International and Indian Studies

Multiple studies have identified practical hurdles in retrofitting:

1. Thermal Management

EV motors and batteries require controlled temperatures. Retrofitted vehicles often lack airflow pathways or cooling plates designed for high-voltage components.

2. Battery Packaging & Structural Constraints

Placing battery packs without compromising crash safety is a major challenge. Literature recommends chassis reinforcement, underbody protection plates, and modular distribution.

3. Weight Distribution

Adding a large battery in the rear can cause imbalance, affecting braking stability and steering response.

4. High-Voltage Safety

Retrofitted vehicles face risks related to:

- improper insulation
- exposed terminals
- inadequate fusing
- poor connector selection

These can lead to electrical fires or short circuits.

5. Regulatory Restrictions

In India, lack of clear guidelines for registration, insurance, and certification has slowed retrofitting adoption.

These challenges form the basis for future innovations and research directions.

2.13 Economic and Environmental Evaluations of Retrofitting

Several studies evaluate the financial and environmental impact of retrofitting:

Economic Findings

- Retrofit cost is 30–60% cheaper than buying a new EV.
- Operating cost is reduced by around 70–80% compared to petrol vehicles.
- Fleet operators recover retrofit investment within 18–30 months through fuel savings alone.

Environmental Findings

- Retrofitting reduces lifecycle carbon emissions by 40–60% compared to manufacturing a new car.
- Studies show that each retrofitted vehicle reduces 2–4 tons of CO_2 annually.
- Reusing chassis and metals reduce the energy-intensive recycling processes.

This makes retrofitting both economically viable and environmentally superior, especially for countries with large populations of ageing ICE vehicles.

2.14 Emerging Research Themes and Future Technologies

Modern research in EV retrofitting focuses on several cutting-edge themes:

A. Second-Life EV Batteries

Batteries retired from commercial EVs (e.g., Nexon EV, MG ZS) still retain 70–80% capacity. Researchers propose using them for retrofits to reduce cost while promoting circular economy.

B. Solar-Assisted Retrofit Systems

Studies explore integrating flexible solar panels on roofs to extend range by 8–12 km/day in Indian sunlight conditions.

C. AI-Based Energy Management

Machine learning can improve range predictions, optimize power consumption, and enhance regenerative braking behavior.

D. Smart Chargers and V2G (Vehicle-to-Grid)

Future retrofits may support:

- bidirectional charging
- grid stabilization
- emergency home power supply

E. Modular "Plug-and-Play" Retrofit Kits

Universities and startups are developing standardized retrofit kits with:

- universal motor mounts
- matched controllers
- configurable battery modules
- easy homologation

This can drastically reduce installation time and cost.

CHAPTER 3: METHODOLOGY

The methodology of this project outlines the systematic approach adopted for studying, analysing, and partially implementing the conversion of an Internal Combustion Engine (ICE) car into an Electric Vehicle (EV). The process was divided into sequential stages that reflect real-world retrofit practices, while also aligning with the academic requirements of modelling, theoretical calculations, simulation, and validation. The methodology integrates mechanical, electrical, and computational domains to derive an engineering-accurate understanding of the retrofit process.

3.1 Overall Project Workflow

The project follows a structured engineering workflow:

1. **Problem Understanding and Requirement Analysis:** Identifies the core challenge of converting an ICE vehicle to an EV affordably and defined the technical, economic, and environmental objectives of the project.
2. **Physical Study of Existing ICE Vehicle (Maruti 800):** Examines the vehicle’s architecture, identified removable ICE components, and assessed available space for mounting EV components.
3. **Component Removal Plan & System Architecture Redesign:** Maps each removed ICE component to its EV equivalent and redesigned the overall powertrain architecture around a single-speed electric drivetrain.
4. **Theoretical Calculations (Forces, Power, Energy):** Derived rolling resistance, aerodynamic drag, and gradient forces to determine the minimum motor power and battery energy required for the target performance.
5. **Selection of Motor, Controller, and Battery Specifications**
6. **MATLAB/Simulink Vehicle Model Implementation:** Built the complete BEV simulation model in Simscape by configuring all subsystems with the Maruti 800's calculated parameters.

7. **Simulation of Vehicle Performance:** Ran the configured model through a representative urban drive cycle and recorded battery, motor, and vehicle behavior across both output scopes.
8. **Interpretation and Validation of Results**
9. **Cost and Feasibility Analysis**

This staged approach ensures that every engineering decision is backed by both calculations and simulation.

3.2 Study of the Existing Vehicle Layout

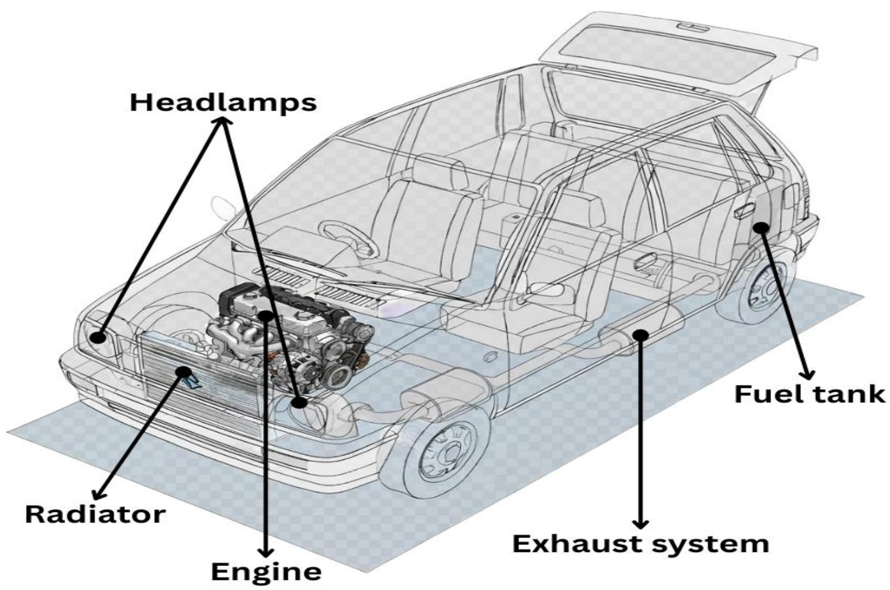

Figure 3.1 Car Layout Configuration

The first step was to understand the complete architecture of the donor vehicle   a Maruti 800. The following components were identified as removable:

- Internal combustion engine
- Fuel tank

- Exhaust system
- Radiator and cooling system
- Air intake and filter
- Carburettor / fuel injectors
- Clutch system (in direct-drive conversions)
- Associated wiring, sensors, and hoses
- The removed mass (~180 kg based on vehicle data) plays a crucial role in recalculating the required power and energy.

The available mounting points, drivetrain path, and space for battery modules were examined. Common feasible locations include:

- Engine bay (primary mounting area)
- Under the rear seat
- Boot space
- Chassis frame reinforcements

This spatial understanding guided the selection of battery size and motor mounting geometry.

3.3 Development of EV Architecture

The EV architecture was designed to replace ICE subsystems with electrical subsystems.

ICE Components → EV Components Mapping

Table 1. Removed and added component

Removed ICE Component	Added EV Component
Engine	PMSM Electric Motor (10 KW)
Fuel Tank	Lithium Battery Pack (10.24 kWh)

Radiator	Motor Controller + DC-DC Converter
Exhaust	Motor Mount + Wiring Harness
Alternator	DC-DC Converter (48V → 12V)

The architecture follows a **simple single-speed reduction gear design**, suitable for low-power small cars.

Figure 3.2 Car after disassembling engine component

3.4 Theoretical Calculations

To correctly size the motor and battery, the primary forces acting on the vehicle were derived:

Rolling Resistance (Fr)

Drag Force (Fd)

Gradient Force (Fg)

Rolling Resistance (Fr):

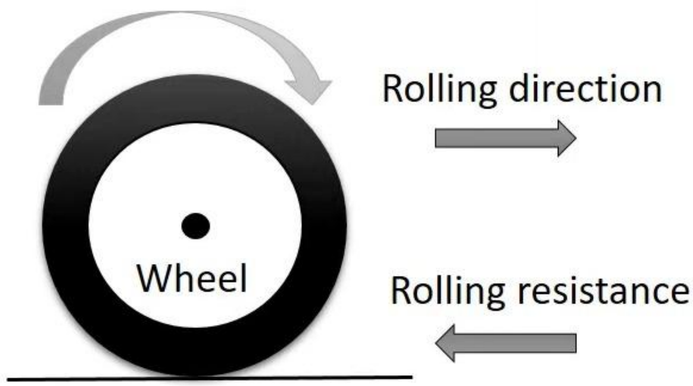

Figure 3.3 Rolling Resistance

Rolling resistance is the resistive force acting opposite to the direction of motion due to deformation of the tire, road surface, and internal hysteresis losses. When a tire rolls, the contact patch deforms. This shifts the normal reaction force slightly ahead of the wheel center. Because of this shift, a small torque is generated which opposes motion.

Derivation

Let:

- $W = mg$= Weight of vehicle
- a= horizontal displacement of normal reaction from wheel center
- R= wheel radius
- F_r= rolling resistance force

The resisting torque due to deformation is:

$$T = W \cdot a$$

This torque must be balanced by the horizontal resisting force:

$$T = F_r \cdot R$$

Thus,

$$F_r R = Wa$$
$$F_r = W \cdot \frac{a}{R}$$

Define the coefficient of rolling resistance:

$$\mu_r = \frac{a}{R}$$

So, the rolling resistance force becomes:

$$\boxed{F_r = \mu_r mg}$$

$\boldsymbol{F_r}$= μ × M × g

where

μ = rolling resistance coefficient

M = mass of vehicle

g = acceleration due to gravity

Given

μ = 0.02, M= 850 kg, g = 9.81

$\boldsymbol{F_r}$ = 0.02 x 850 x 9.81 = 166.77 N

DRAG FORCE (Fd):

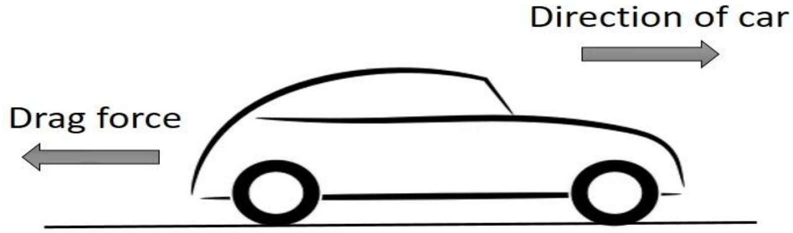

Figure 3.4 Drag Force

Drag force is produced by air resistance as the vehicle moves forward. From fluid mechanics, drag force on a body is:

$$F_d = \frac{1}{2}\rho V^2 C_d A$$

Derivation

The kinetic pressure of airflow is:

$$P = \frac{1}{2}\rho V^2$$

Where,

- ρ= air density
- V= relative velocity of air

The drag force is this pressure multiplied by the effective frontal area $A$and scaled by drag coefficient C_d:

$$F_d = P \cdot C_d \cdot A$$

Substituting:

$$F_d = (\frac{1}{2}\rho V^2)C_d A$$

Thus:

$$\boxed{F_d = \frac{1}{2}\rho C_d A V^2}$$

F_d=0.5× ρ × Cd × A_f × V^2

ρ = density of air (kg/m^3)
Cd = coefficient of drag
A = vehicle front area (m^2)
V = mean vehicle velocity (m/s)

ρ = 1.2 kg/ m^3**,** Cd = 0.34, A_f = 1.44 x 1.405 = 2.026 m^2

Effective A_f = 2.026 x 0.8 = 1.62 m^2

V = 18.05 m/s (65kmph)

F_d = 0.5 x 1.2 x 0.34 x 1.62 x $(18.05)^2$ = 107.73 N

GRADIENT FORCE (F_g):

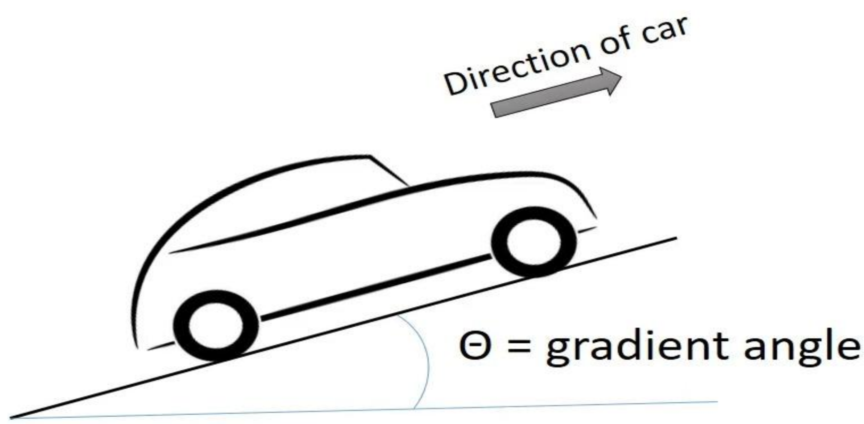

Figure 3.5 Gradient Force

Gradient force (also called grade resistance or hill-climb force) is the component of the vehicle's weight acting **down the slope**. When a vehicle moves uphill at an incline θ, its weight $W = mg$ acts vertically downward.

Resolve weight into components:

- Perpendicular to slope: *mgcosθ*
- Parallel to slope: *mgsinθ* → **opposes motion**

Derivation

Parallel component resisting motion:

$$\boxed{F_g = mg\sin\,\theta}$$

This is the gradient force required to keep the vehicle moving uphill.

$\boldsymbol{F_g}$ = m x g x sinθ

where:

m = mass of vehicle

g = acceleration due to gravity

θ = gradient angle

$\boldsymbol{F_g}$ = 850 x 9.81 x sin5 = 726.74 N

Total Traction Force

F =$\boldsymbol{F_r}$+ $\boldsymbol{F_d}$ + $\boldsymbol{F_g}$

F = 166.77 + 107.73 + 726.74 = 1001.24 N

Motor Power

Power required is the rate at which work is done to overcome resisting forces.

Work done = Force × distance

Power = Work / time

Thus:

$$P = \frac{F \cdot s}{t} = F \cdot \frac{s}{t}$$

$$\boxed{P = F \cdot V}$$

Since $V = s/t$

Where:

- F= total tractive force (rolling + drag + gradient)
- V= vehicle velocity

This is the core EV powertrain sizing formula.

Motor Power = Force x Velocity

= 1001.24 x 18.05 = 18.07 kw (peak power)

Rated Motor Power = 18.07/3 = 6.0241 kw = 7 kw (approx)

Battery Energy

Energy required for a trip is:

$$\text{Work} = \text{Force} \times \text{Distance}$$

Let:

- $F_t = F_r + F_d$(average resistive forces on flat road)
- D =distance in meters

So:

$$W = F_t \cdot D$$

To convert Joules → kWh:

$$E_{\text{kWh}} = \frac{F_t \cdot D}{3.6 \times 10^6}$$

Since 1 kWh $= 3.6 \times 10^6$ J

For a distance of 100 km:

$$D = 100 \times 1000 = 100{,}000 \text{ m}$$

Thus:

$$\boxed{E_{\text{kWh}} = \frac{F_t \cdot D}{3600}}$$

Battery Energy = Force x Range / 3600

Force = $\boldsymbol{F_r}$ + $\boldsymbol{F_d}$ = 274.5 N, Range = 100 km

(Gradient force is excluded as we don't run the car continuously on the slope/gradient)

= 274.5 x 100 / 3600 = 7.625 kwh

3.4.1 Actual Calculation

Actual weight = 670 kg, IC parts weight = 170kg; passenger weight = 350kg (5 passengers of 70kg); EV parts = ~100kg (Motor + Battery + Wiring, etc.)

Total weight = 670 - 170 + 350 + 100 = 950 kg

Gravitational acceleration (g) = 9.81m/s²

Rolling resistance coefficient (Cr) = 0.02

Drag coefficient (Cd) = 0.35

Frontal area(car) (a) = 1.8m²

Air density (dry air) (ρ) = 1.2 kg/m³

Vehicle speed (V) = 60 Km/h = 16.67 m/s

Rolling resistance:

F_r = C_r × m_{eff} × g × cos(θ)

Where:

C_r = 0.0156 (speed-dependent rolling resistance coefficient)

m_{eff} = 1004.4 kg (effective vehicle mass with rotational inertia)

g = 9.81 m/s²

θ = gradient angle

Flat Road (θ= 0°)

F_r = 0.0156 × 1004.4 × 9.81 × cos(0°)

F_r = 0.0156 × 1004.4 × 9.81 × 1.0000

F_r = 153.73 N

5° Slope

F_r = 0.0156 × 1004.4 × 9.81 × cos(5°)

F_r = 153.73 × 0.9962 Fr = 153.14 N

10° Slope

F_r = 0.0156 × 1004.4 × 9.81 × cos(10°)

F_r = 153.73 × 0.9848

F_r = 151.40 N

Aerodynamic Drag

F_d = 0.5 × ρ × C_d× A_f × V²

Where:

ρ = 1.136 kg/m³ (air density at 38°C — Hyderabad summer)

C_d = 0.34 (drag coefficient, Maruti 800 — ResearchGate CFD study)

A_f = 1.70 m² (frontal area: 1.44 × 1.405 × 0.84 EcoModder correction)

V = 16.67 m/s (60 km/h)

F_d = 0.5 × 1.136 × 0.34 × 1.70 × (16.67) ²

F_d = 0.5 × 1.136 × 0.34 × 1.70 × 277.89

F_d = 0.5 × 182.58

F_d = 91.3 N

Gradient Force (F_g)

F_g= m_{eff} × g × sin(θ)

Where: m_{eff} = 1004.4 kg, g = 9.81 m/s²

θ = gradient angle

Flat Road (θ = 0°)

F_g = 1004.4 × 9.81 × sin (0°) = 0 N

5° Slope

F_g = 1004.4 × 9.81 × sin (5°)

F_g = 9853.2 × 0.08716

F_g = 859.0 N

10° Slope

F_g = 1004.4 × 9.81 × sin (10°)

F_g = 9853.2 × 0.17365

F_g = 1710.5 N

Total Road Load Force & Wheel Power Required

F_t= Fr + Fd + F_g

P_{wheel}= F_t × V

Flat Road

F_t = 153.73 + 91.3 + 0 = 245.0 N

P_{wheel} = 245.0 × 16.67 = 4085W = 4.09 kW

5° Slope

F_t = 153.14 + 91.3 + 859.0 = 1103.4 N

P_{wheel} = 1103.4 × 16.67 = 18,394W = 18.39 kW

10° Slope

F_t = 151.40 + 91.3 + 1710.5 = 1953.2 N

P_{wheel} = 1953.2 × 16.67 = 32,561W = 32.56 kW

Motor Shaft Power

P_{motor} = P_{wheel} / η_{dt}

η_{dt} = 0.88 (gearbox + differential — standard per SAE/automotive textbooks)

Flat Road

P_{motor} = 4.09 / 0.88 = 4.65 kW

5° Slope

P_{motor}= 18.39 / 0.88 = 20.90 kW

10° Slope

P_{motor} = 32.56 / 0.88 = 36.99 kW

Electrical Power Draw

$P_{electrical}$ = P_{motor} / η_{motor}

$P\eta_{motor}$ = 0.90 (PMSM motor efficiency at rated operating point)

Flat Road

$P_{electrical}$ = 4.65 / 0.90 = 5.17 kW

5° Slope

$P_{electrical}$ = 20.90 / 0.90 = 23.22 kW

10° Slope

$P_{electrical}$ = 36.99 / 0.90 = 41.1 kW

Energy Required for Target Range

Time for 100 km at 60 km/h:

t = 100 / 60 = 1.667 hours

$E_{required} = P_{electrical} \times t$

$E_{required}$ = 5170 W × 1.667 h

$E_{required}$ = 8618 Wh

$E_{required}$= 8.62 kWh

Table 2. Comparison between theoretical and actual calculations

PARAMETER		THEORETICAL	ACTUAL
Weight (Kg)		850	950
Rolling Resistance (F_r)(N)	θ = 0°	166.77	153.73
	θ = 5°	166.77	153.14
	θ = 10°	166.77	151.40
Drag Forces (F_d)(N)	θ = 0°	91.8	91.3
	θ = 5°	91.8	91.3

	θ = 10°	91.8	91.3
Gradient Force (F_g)(N)	θ = 0°	0	0
	θ = 5°	726.74	859.0
	θ = 10°	726.74	1710.5
Motor Power(kW)		6.0241 ~ 6.1	6.6
Battery Energy(kWh)		7.625	8.62

3.5 MATLAB/Simulink Vehicle Model Development

To validate theoretical results, a complete EV simulation model was implemented using the **Simscape Battery Electric Vehicle (BEV) template**.

3.5.1 BEV Simulink Model Overview:

The top-level BEV model represents the entire powertrain and control architecture of an electric vehicle. It begins with the driver and controller subsystem, which generates torque commands based on accelerator and brake inputs. These commands flow into the motor drive unit, which converts electrical power into mechanical torque. The torque then passes through the reduction gear and into the longitudinal vehicle model, which determines vehicle motion.

Electrical power flows in the opposite direction: the high-voltage battery supplies DC power to the motor drive unit. As the motor consumes power, the battery monitors current, voltage, and power, allowing the model to replicate real energy consumption behaviour.

Important subsystems inside the BEV model include:

- Controller & environment
- Battery
- Motor drive unit
- Reduction gear
- Vehicle longitudinal dynamics
- Measurement and feedback blocks

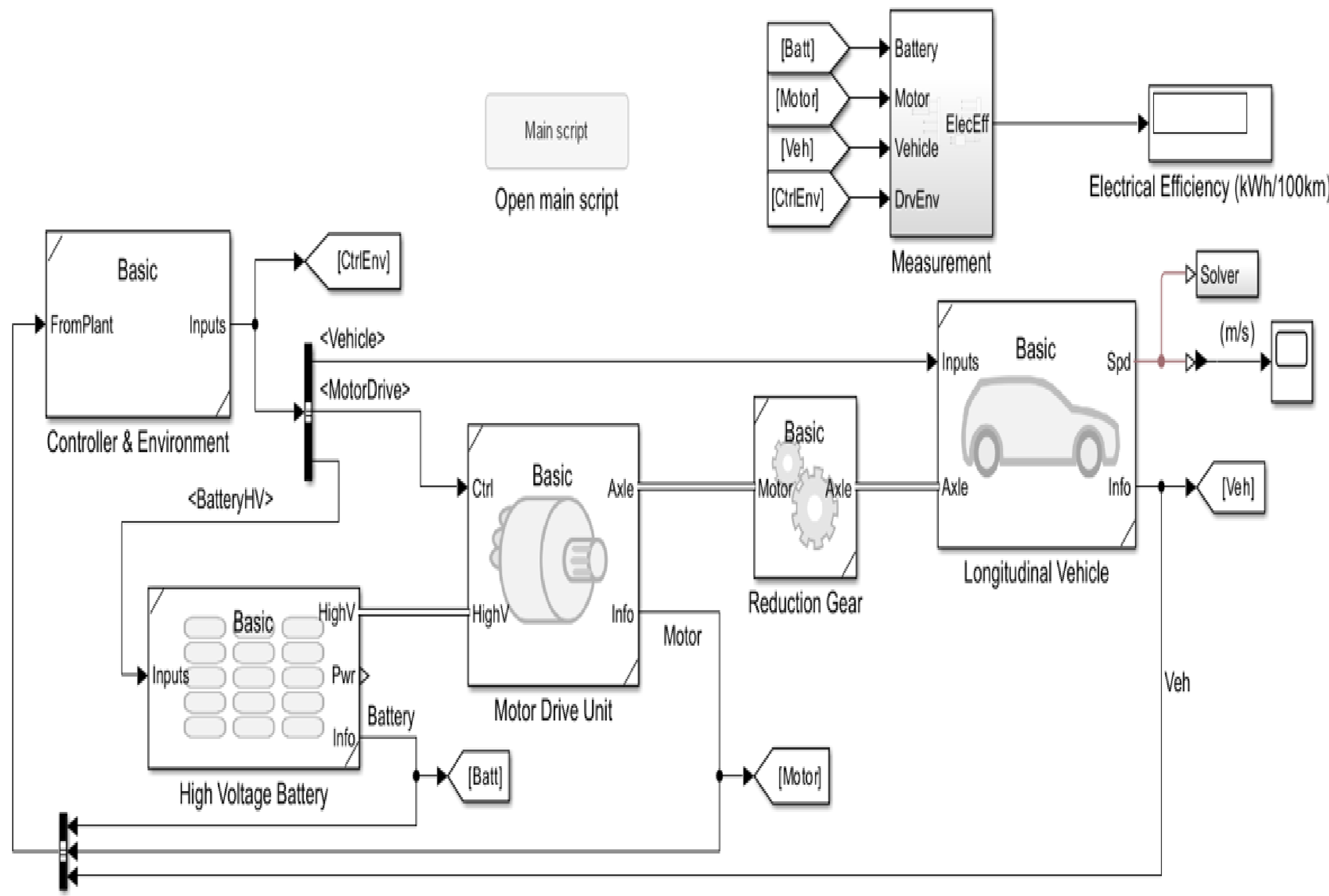

Figure 3.6 Simulink BEV model

3.5.2 Motor Drive Unit:

The Motor Drive Unit (MDU) is the central subsystem that bridges the electrical and mechanical domains of the BEV simulation model. It receives two primary inputs, DC electrical power from the battery pack and a torque command signal from the PI controller and converts them into mechanical shaft rotation that ultimately drives the vehicle. Onboard electrical sensors continuously measure incoming battery voltage and current in real time, allowing the model to accurately compute instantaneous power draw and maintain reliable energy accounting across the entire simulation. A torque limiting stage ensures the motor never exceeds its physical design ceiling, preventing the controller from commanding forces beyond what a real 10 kW PMSM motor could physically produce.

The mechanical behaviour of the motor is modelled through three internal elements rotor inertia, a rotor damper, and a torsional spring-damper. The rotor inertia of 0.002 $kg \cdot m^2$ ensures the motor shaft cannot change speed instantaneously, replicating the gradual acceleration and deceleration of a real motor. The rotor damper models viscous friction losses from bearings and windage, while the torsional spring-damper acts as a compliance buffer between the motor output and the downstream drivetrain. This spring-damper is particularly important because it absorbs sudden torque changes and prevents them from propagating instantaneously through the system, which would otherwise cause algebraic constraint violations and numerical instability in the Simscape solver.

Motor efficiency is applied at the rated operating point of 3500 RPM and 40 Nm, with a combined motor and driver efficiency of 88%. This means 12% of all electrical power drawn from the battery is lost as heat through iron losses, fixed losses, and copper losses in the windings — ensuring the simulation correctly reflects the fact that electrical power consumed is always greater than mechanical power delivered. The final output of the MDU is a rotational Simscape bus carrying motor speed and torque to the Reduction Gear subsystem, while simultaneously feeding back into the measurement blocks for scope display and closed-loop PI control. Its correct configuration was essential to producing the stable, physically consistent simulation results observed across both output scopes.

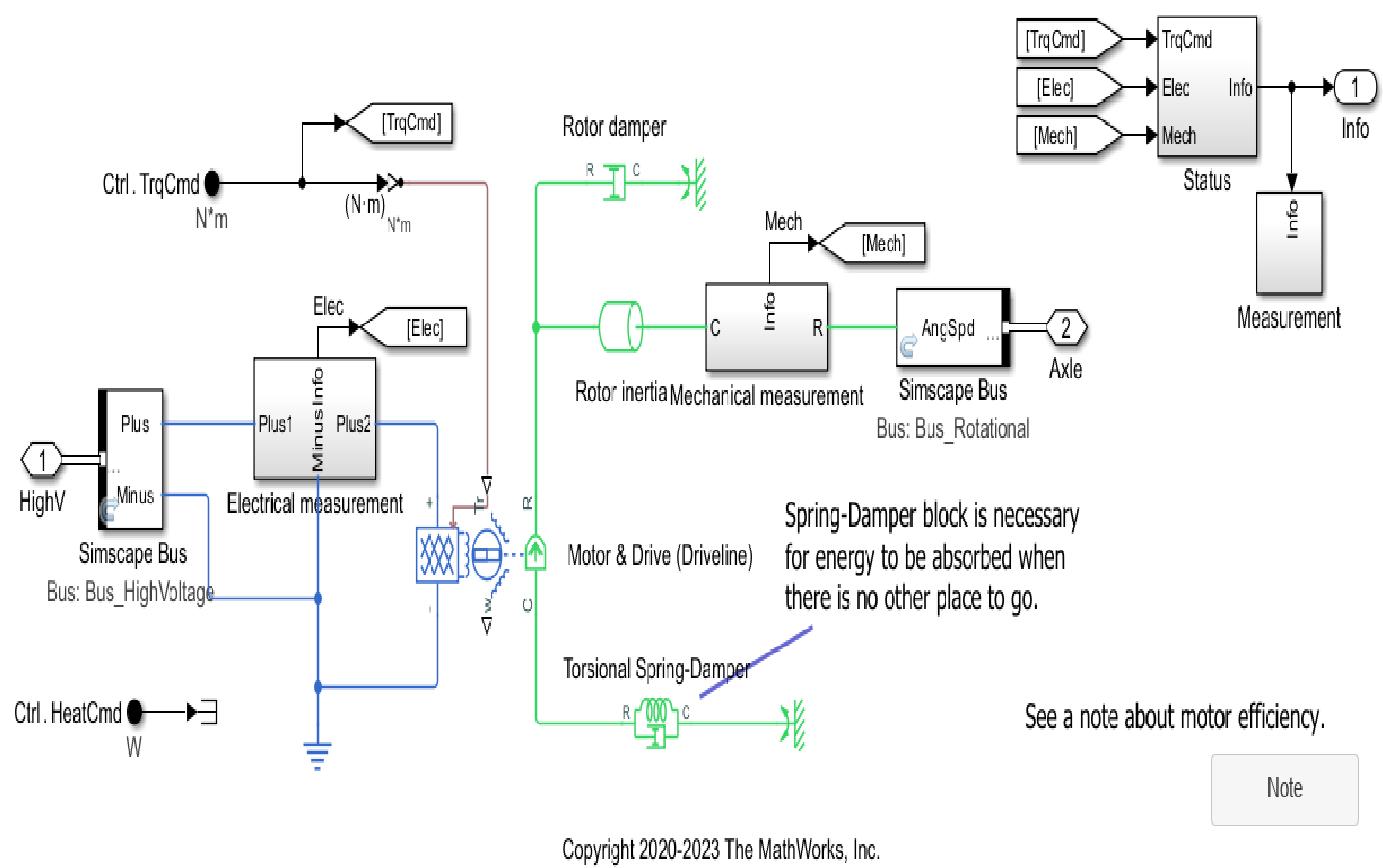

Figure 3.7 Motor model

3.5.3 High-Voltage Battery Unit:

The High Voltage Battery subsystem serves as the primary electrical energy source for the entire BEV simulation model. At its core, it contains a DC voltage source representing the nominal pack voltage of the lithium-ion battery selected for the Maruti 800 retrofit. Connected in series with this voltage source is an internal resistance, which is a critical modelling element that causes realistic voltage sag under high current demand. When the motor draws large currents during acceleration, this internal resistance produces a proportional voltage drop, bringing the terminal voltage down from its nominal value which is exactly consistent with real lithium-ion battery behaviour under load. This combination of voltage source and internal resistance ensures that the battery does not behave as an ideal unlimited power supply, but instead responds realistically to the varying electrical demands placed on it throughout the drive cycle.

The subsystem is equipped with dedicated electrical sensors that continuously measure three key quantities, battery terminal voltage, current flowing in and out of the pack, and instantaneous electrical power being consumed or recovered. These measurements are not just passive observations; they are actively used by the rest of the model to compute state of charge depletion, energy consumption, and electrical efficiency in real time. All of these signals are organized and routed through a Simscape high-voltage bus, a structured data pathway that carries the complete battery status information to every downstream subsystem that needs it, including the Motor Drive Unit, the Measurement blocks, and the output scopes. A pair of Bus Creator blocks within the subsystem further organizes these electrical and measurement outputs into clean, structured signals that are easy for downstream subsystems to access and process.

The battery subsystem is also configured to support bidirectional current flow, which is essential for modelling regenerative braking. During deceleration, when the motor operates as a generator and pushes current back toward the battery, the subsystem correctly accepts this reverse current and reflects it as a rise in terminal voltage above the nominal value which is the standard open-circuit voltage recovery behaviour observed in real lithium-ion cells during charging. The simulation correctly tracks the gradual depletion of charge throughout the drive cycle, with a small recovery visible during the regenerative braking phase.

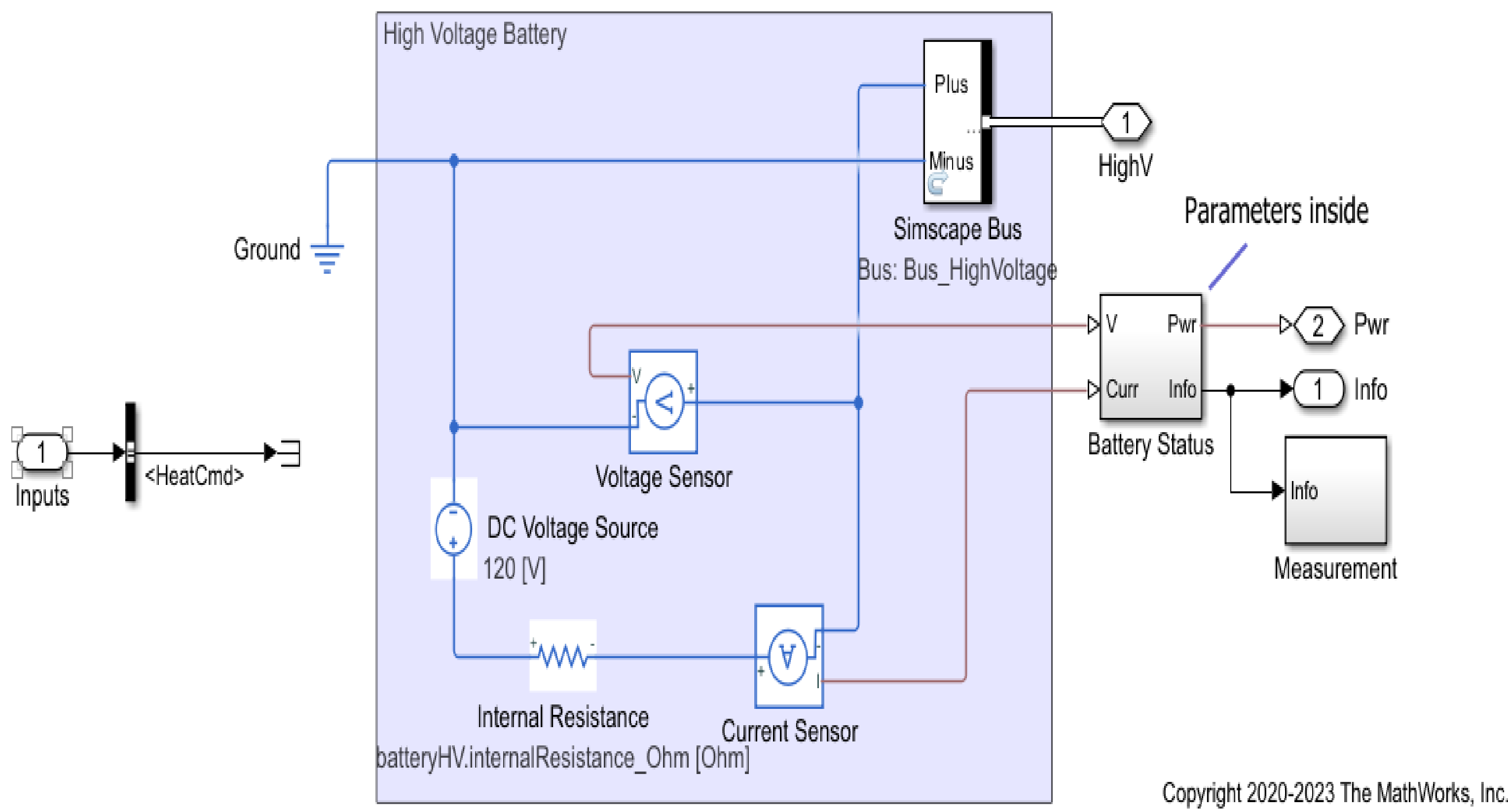

Figure 3.8 Battery model

3.6 Retrofit Component Selection Logic

- **Motor**: Chosen to match tractive force requirements and acceleration profile.
- **Battery**: Capacity calculated using energy per km and range needs.
- **Controller**: Selected to match motor’s voltage and current requirements with programmable regen.
- **Mounting Framework**: Engine bay adapted to hold motor + mounts.
- **Wiring Harness**: High-voltage cables routed with insulation and protective conduits.
- **Safety Systems**: Fuses, relays, and emergency shut-off switch planned.

3.7 MATLAB Programming

Editor - C:\Users\FAIZ MUBARAK\Downloads\project images\Simscape-Battery-Electric-Vehicle-Model-R2023a\Simscape-Battery-Electric-Vehicle-Model-R2023a\BEV_MASTER.m

BEV_MASTER.m | untitled

```
%% BEV_MASTER.m
%  THE ONLY SCRIPT YOU NEED.
%  Run this once after opening the project. It fixes everything:
%    1. Clears any broken callback
%    2. Loads all Maruti 800 parameters
%    3. Sets smooth drive cycle (start=0, no spikes)
%    4. Fixes torque integrator limit
%    5. Sets solver
%    6. Installs a correct permanent callback
%    7. Runs the simulation

modelName = 'BEV_system_model';
if ~bdIsLoaded(modelName); open_system(modelName); pause(3); end

%% STEP 1: Clear broken callback immediately
set_param(modelName, 'InitFcn', '');
fprintf('[1] Cleared old callback\n');

%% STEP 2: Load all Maruti 800 parameters into workspace
defineBus_HighVoltage;
defineBus_Rotational;

vehicle.mass_kg               = 950;
vehicle.tireRollingRadius_m   = 0.272;
vehicle.tireRollingCoeff      = 0.0156;
vehicle.airDragCoeff          = 0.34;
vehicle.frontalArea_m2        = 1.70;
vehicle.gravAccel_m_per_s2    = 9.81;

smoothing.vehicle_speedThreshold_kph     = 1;
smoothing.vehicle_axleSpeedThreshold_rpm = 1;
smoothing.Reducer_PowerThreshold_W       = 1;

```

Figure **3.9 MATLAB Script**

Simulation Setup and Execution: BEV_MASTER.m

The MATLAB script BEV_MASTER.m serves as the central configuration and execution file for the Battery Electric Vehicle (BEV) simulation. It initialises the Simulink model, loads all vehicle parameters, defines the drive cycle, configures the solver, and runs the simulation in a single execution. The script is structured into seven clearly defined steps, described below.

Step 1 — Clearing Stale Callbacks

The script begins by verifying whether the Simulink model BEV_system_model is already loaded; if not, it opens the model and waits three seconds for it to fully initialise. It then clears any previously registered InitFcn callback on the model. This prevents residual or broken callbacks from earlier sessions from interfering with the current run.

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BEV_MASTER.m

```
batteryHV.nominalVoltage_V    = 48;
batteryHV.nominalCapacity_kWh = 9.6;
batteryHV.nominalCharge_Ahr   = BatteryHV_getAmpereHourRating( ...
    Voltage_V=48, Capacity_kWh=9.6, StateOfCharge_pct=100);
batteryHV.internalResistance_Ohm        = 0.005;
batteryHV.ThermalMass_J_per_K           = 1600;
batteryHV.measuredVoltage_V             = 43.2;
batteryHV.measuredCharge_Ahr            = batteryHV.nominalCharge_Ahr * 0.5;
batteryHV.measurementTemperature_K      = 311.15;
batteryHV.secondMeasurementTemperature_K= 273.15;
batteryHV.secondNominalVoltage_V        = 45.6;
batteryHV.secondInternalResistance_Ohm  = 0.01;
batteryHV.secondMeasuredVoltage_V       = 43.2;
batteryHV.ambientTemp_K                 = 311.15;
batteryHV.ambientMass_t                 = 10000;
batteryHV.ambientSpecificHeat_J_per_Kkg = 1000;
batteryHV.RadiationArea_m2              = 1;
batteryHV.RadiationCoeff_W_per_K4m2    = 5e-10;

motorDriveUnit.trqMax_Nm                = 48;
motorDriveUnit.powerMax_kW              = 12;
motorDriveUnit.responseTime_s           = 0.02;
motorDriveUnit.rotorInertia_kg_m2       = 0.002;
motorDriveUnit.rotorDamping_Nm_per_radps= 1e-4;
motorDriveUnit.efficiency_pct           = 88;
motorDriveUnit.spd_eff_rpm              = 3500;
motorDriveUnit.trq_eff_Nm              = 40;
motorDriveUnit.ironLoss_W               = 50;
motorDriveUnit.fixedLoss_W              = 20;
motorDriveUnit.ThermalMass_J_per_K      = 800;
motorDriveUnit.ambientTemp_K            = 311.15;
motorDriveUnit.ambientMass_t            = 10000;
```

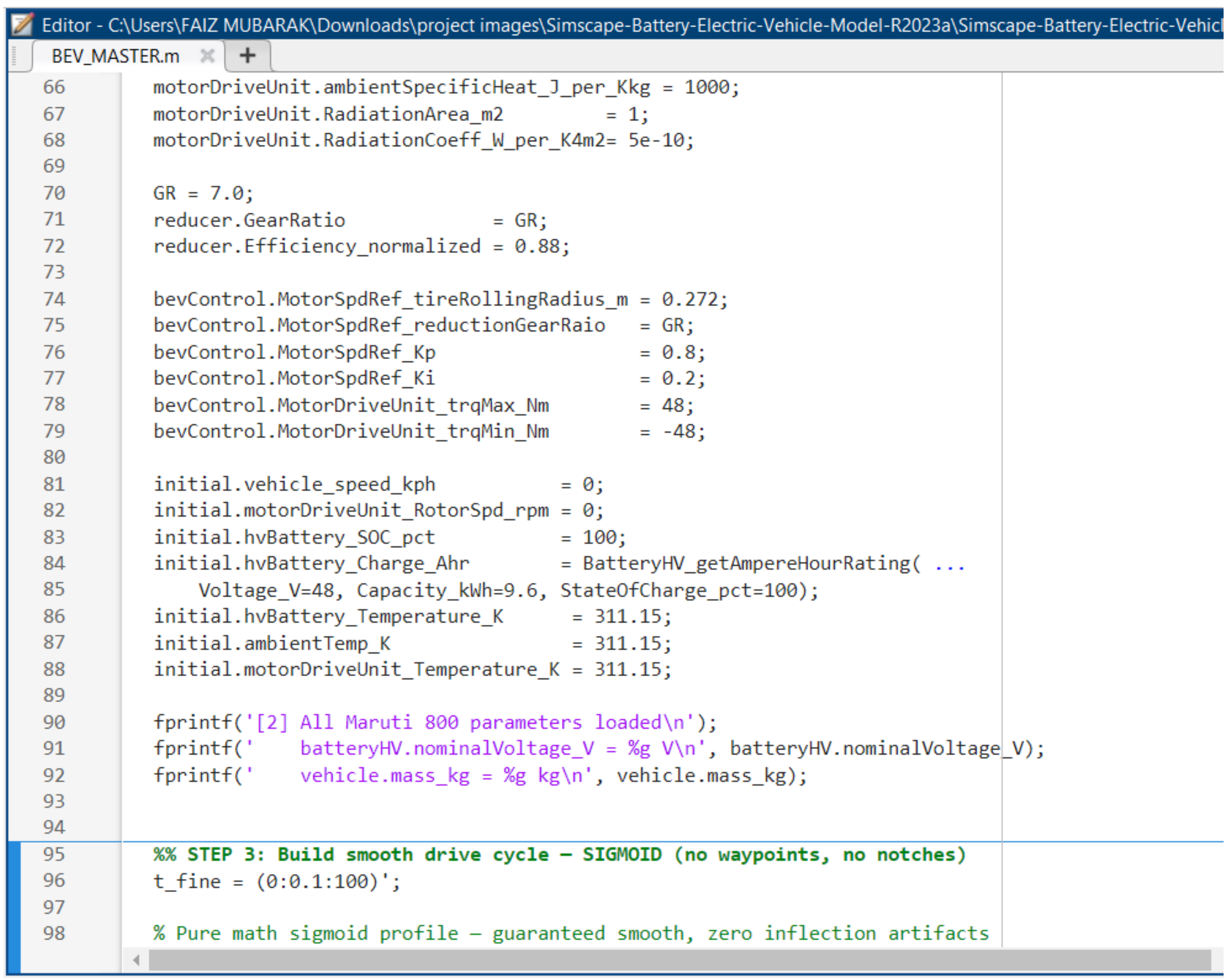
Figure **3.10 MATLAB Script**

Step 2 — Vehicle and System Parameter Initialisation

All physical and electrical parameters of the simulated vehicle — modelled after the Maruti 800 — are loaded into the MATLAB workspace. These are organised into structured variables:

Table 3: Vehicle Dynamics

Parameter	Value
Kerb Mass	950 kg
Tyre Rolling Radius	0.272 m
Rolling Resistance Coefficient	0.0156
Aerodynamic Drag Coefficient	0.34
Frontal Area	1.70 m²



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BEV_MASTER.m

```
motorDriveUnit.ambientSpecificHeat_J_per_Kkg = 1000;
motorDriveUnit.RadiationArea_m2          = 1;
motorDriveUnit.RadiationCoeff_W_per_K4m2= 5e-10;

GR = 7.0;
reducer.GearRatio             = GR;
reducer.Efficiency_normalized = 0.88;

bevControl.MotorSpdRef_tireRollingRadius_m = 0.272;
bevControl.MotorSpdRef_reductionGearRaio   = GR;
bevControl.MotorSpdRef_Kp                  = 0.8;
bevControl.MotorSpdRef_Ki                  = 0.2;
bevControl.MotorDriveUnit_trqMax_Nm        = 48;
bevControl.MotorDriveUnit_trqMin_Nm        = -48;

initial.vehicle_speed_kph          = 0;
initial.motorDriveUnit_RotorSpd_rpm = 0;
initial.hvBattery_SOC_pct          = 100;
initial.hvBattery_Charge_Ahr       = BatteryHV_getAmpereHourRating( ...
    Voltage_V=48, Capacity_kWh=9.6, StateOfCharge_pct=100);
initial.hvBattery_Temperature_K     = 311.15;
initial.ambientTemp_K               = 311.15;
initial.motorDriveUnit_Temperature_K = 311.15;

fprintf('[2] All Maruti 800 parameters loaded\n');
fprintf('    batteryHV.nominalVoltage_V = %g V\n', batteryHV.nominalVoltage_V);
fprintf('    vehicle.mass_kg = %g kg\n', vehicle.mass_kg);


%% STEP 3: Build smooth drive cycle — SIGMOID (no waypoints, no notches)
t_fine = (0:0.1:100)';

% Pure math sigmoid profile — guaranteed smooth, zero inflection artifacts
```

Figure **3.11 MATLAB Script**

High-Voltage Battery (batteryHV)

The battery is modelled as a 48 V, 9.6 kWh pack. Key parameters include internal resistance (0.005 Ω), thermal mass (1600 J/K), initial state of charge (100%), and both primary and secondary measurement points for voltage and temperature — enabling a two-point battery characterisation model. The nominal charge in ampere-hours is computed dynamically using the helper function BatteryHV_getAmpereHourRating().

Motor Drive Unit (motorDriveUnit)

The electric motor is rated at 48 Nm peak torque and 12 kW peak power, with a response time of 0.02 s. Efficiency is specified at 88% at a reference operating point of 3500 rpm / 40 Nm. Iron and fixed losses (50 W and 20 W respectively) are included for thermal modelling.

Reducer (Gearbox)

A single fixed-ratio reducer is used with a gear ratio of 7.0 and a transmission efficiency of 88%.

BEV Controller (bevControl)

The controller parameters include the tyre rolling radius and gear ratio (used internally for speed reference conversion), and the proportional–integral (PI) gains for the speed controller (Kp = 0.8, Ki = 0.2 in the main script). Torque output is clamped symmetrically to ±48 Nm.

Initial Conditions

All states are initialised to rest: vehicle speed = 0 km/h, rotor speed = 0 rpm, battery SOC = 100%, and all temperatures set to 311.15 K (38°C), representing an ambient Hyderabad-like thermal condition.

Step 3 — Drive Cycle Definition

A smooth velocity reference profile is generated mathematically using a sigmoid function, avoiding the discontinuities and derivative spikes that arise from piecewise linear (waypoint) profiles.

The profile is defined over a 100-second window at 0.1-second resolution. Two opposing sigmoids are combined to produce a smooth trapezoidal-like speed trace that rises to 60 km/h and falls back to 0:

rise = 60 / (1 + exp(−0.25·(t − 25)))

fall = 60 / (1 + exp(0.25·(t − 75)))

v(t) = max(rise + fall − 60, 0)

The profile is clamped to zero to prevent any negative velocity artefacts. From this velocity trace, the acceleration profile is derived using a numerical gradient (divided by 3.6 for unit conversion from km/h to m/s^2). A constant gear signal of 1 is used throughout, consistent with the single-speed drivetrain.

The computed data is written both to workspace variables (DriveCycle, AccelCycle, GearCycle) and directly into the Simulink 1-D Lookup Table block within the model hierarchy, ensuring that the model and workspace remain synchronised.

Step 4 — Controller and Input Block Configuration

Two corrections are applied via set_param calls:

1. Torque Integrator Saturation Limits — The speed controller's integrator saturation block (labelled Limits [-40,40]) is updated to ±48 Nm, matching the motor's actual torque rating. This prevents integrator windup beyond the physical capability of the drive unit.
2. Drive Cycle Step Block — A Step block within the drive pattern sub-system is configured to activate at t = 0 s, with an initial value of 0 and a final value of 1. This ensures the drive cycle reference is active from the start of the simulation with no delay.

Step 5 — Solver Configuration

The Simulink solver is set to ode15s, a variable-order, stiff solver appropriate for systems with mixed fast and slow dynamics — characteristic of combined electrical, mechanical, and thermal sub-systems. The solver settings used are:

Setting	Value
Solver	mode15s

Maximum Step Size	0.05 s
Relative Tolerance	1×10^{-4}
Absolute Tolerance	1×10^{-5}

Table 4: Solver Setting

Step 6 — Permanent Callback Installation

A self-contained InitFcn callback string is constructed programmatically and registered on the model. This callback replicates all parameter loading, drive cycle generation, and solver/block configuration, so that the simulation can be re-run in a future MATLAB session simply by opening and running the model — without needing to manually execute BEV_MASTER.m again. The model is saved after the callback is installed to persist this configuration.

Note that the callback uses a piecewise PCHIP-interpolated drive cycle (waypoints at t = 0, 5, 30, 60, 90, 100 s), which is a slightly different formulation from the sigmoid used in the main script body, but produces a comparable smooth profile.

Step 7 — Simulation Execution

Finally, the script calls sim(modelName) to run the Simulink model with all configured parameters and initial conditions. Post-simulation, results can be viewed on the Scope HV Battery and Scope Vehicle blocks within the model.

Summary

BEV_MASTER.m provides a single-entry-point workflow for the BEV simulation, ensuring reproducibility by encoding all parameters, drive cycle data, and solver settings in one place. The use of a sigmoid drive cycle avoids numerical issues at simulation start, the ode15s solver handles the stiff multi-domain dynamics robustly, and the permanent callback guarantees that the model remains self-consistent across MATLAB sessions.

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BEV_MASTER.m

```matlab
rise = 60 ./ (1 + exp(-0.25*(t_fine - 25)));   % smooth rise centred at t=25
fall = 60 ./ (1 + exp( 0.25*(t_fine - 75)));   % smooth fall centred at t=75
v_fine = max(rise + fall - 60, 0);              % combine and clamp to 0

% Update all workspace variables
sig.Data.X       = t_fine;
sig.Data.Y       = v_fine;
DriveCycle.Time = t_fine;
DriveCycle.Data = v_fine;
AccelCycle.Time = t_fine;
AccelCycle.Data = gradient(v_fine, 0.1) / 3.6;
GearCycle.Time  = t_fine;
GearCycle.Data  = ones(size(t_fine));

cycleRepeat   = 0;
InitialOutput = 0;
start         = 0;

% Overwrite lookup table inside model
lutPath = [modelName '/Controller & Environment/Vehicle speed reference/Simple drive pattern/1-D Lookup Table'];
try
    set_param(lutPath, ...
        'BreakpointsForDimension1', mat2str(t_fine'), ...
        'Table',                    mat2str(v_fine'));
    fprintf('[3] Sigmoid drive cycle written to model: %d points, max %.0f km/h\n', length(t_fine), max(v_fine));
catch e
    fprintf('[3] Lookup table warn: %s\n', e.message);
end
%% STEP 4: Fix torque integrator limit and Step block via set_param
try
    set_param([modelName '/Controller & Environment/BEV Controller/Speed Controller/Limits [-40,40]'], ...
        'UpperSaturationLimit', '48', 'LowerSaturationLimit', '-48');
    fprintf('[4] Torque integrator limit: [-48, +48] Nm\n');
```

Figure **3.12 MATLAB Script**

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BEV_MASTER.m

```matlab
catch e
    fprintf('[4] Torque limit warn: %s\n', e.message);
end

try
    stepPath = [modelName '/Controller & Environment/Vehicle speed reference/Simple drive pattern/Ramp/Step'];
    set_param(stepPath, 'Time', '0', 'Before', '0', 'After', '1');
    fprintf('[4] Step block: fires at t=0\n');
catch e
    fprintf('[4] Step warn: %s\n', e.message);
end

%% STEP 5: Solver settings
set_param(modelName, 'Solver', 'ode15s', 'MaxStep', '0.05', 'RelTol', '1e-4', 'AbsTol', '1e-5');
fprintf('[5] Solver: ode15s, MaxStep=0.05s\n');

%% STEP 6: Install correct permanent callback (no external file, start=0)
cb = [ ...
    'defineBus_HighVoltage; defineBus_Rotational; ' ...
    'vehicle.mass_kg=950; vehicle.tireRollingRadius_m=0.272; vehicle.tireRollingCoeff=0.0156; ' ...
    'vehicle.airDragCoeff=0.34; vehicle.frontalArea_m2=1.70; vehicle.gravAccel_m_per_s2=9.81; ' ...
    'smoothing.vehicle_speedThreshold_kph=1; smoothing.vehicle_axleSpeedThreshold_rpm=1; smoothing.Reducer_PowerThreshold_W=1; ' ...
    'batteryHV.nominalVoltage_V=48; batteryHV.nominalCapacity_kWh=9.6; ' ...
    'batteryHV.nominalCharge_Ahr=BatteryHV_getAmpereHourRating(Voltage_V=48,Capacity_kWh=9.6,StateOfCharge_pct=100); ' ...
    'batteryHV.internalResistance_Ohm=0.005; batteryHV.ThermalMass_J_per_K=1600; ' ...
    'batteryHV.measuredVoltage_V=43.2; batteryHV.measuredCharge_Ahr=batteryHV.nominalCharge_Ahr*0.5; ' ...
    'batteryHV.measurementTemperature_K=311.15; batteryHV.secondMeasurementTemperature_K=273.15; ' ...
    'batteryHV.secondNominalVoltage_V=45.6; batteryHV.secondInternalResistance_Ohm=0.01; ' ...
    'batteryHV.secondMeasuredVoltage_V=43.2; batteryHV.ambientTemp_K=311.15; ' ...
    'batteryHV.ambientMass_t=10000; batteryHV.ambientSpecificHeat_J_per_Kkg=1000; ' ...
    'batteryHV.RadiationArea_m2=1; batteryHV.RadiationCoeff_W_per_K4m2=5e-10; ' ...
    'motorDriveUnit.trqMax_Nm=48; motorDriveUnit.powerMax_kW=12; motorDriveUnit.responseTime_s=0.02; ' ...
    'motorDriveUnit.rotorInertia_kg_m2=0.002; motorDriveUnit.rotorDamping_Nm_per_radps=1e-4; ' ...
```

Figure **3.13 MATLAB Script**

Editor - C:\Users\FAIZ MUBARAK\Downloads\project images\Simscape-Battery-Electric-Vehicle-Model-R2023a\Simscape-Battery-Electric-Vehicle-Model-R2023a\BEV_MASTER.m

BEV_MASTER.m

```matlab
    'batteryHV.RadiationArea_m2=1; batteryHV.RadiationCoeff_W_per_K4m2=5e-10; ' ...
    'motorDriveUnit.trqMax_Nm=48; motorDriveUnit.powerMax_kW=12; motorDriveUnit.responseTime_s=0.02; ' ...
    'motorDriveUnit.rotorInertia_kg_m2=0.002; motorDriveUnit.rotorDamping_Nm_per_radps=1e-4; ' ...
    'motorDriveUnit.efficiency_pct=88; motorDriveUnit.spd_eff_rpm=3500; motorDriveUnit.trq_eff_Nm=40; ' ...
    'motorDriveUnit.ironLoss_W=50; motorDriveUnit.fixedLoss_W=20; motorDriveUnit.ThermalMass_J_per_K=800; ' ...
    'motorDriveUnit.ambientTemp_K=311.15; motorDriveUnit.ambientMass_t=10000; ' ...
    'motorDriveUnit.ambientSpecificHeat_J_per_Kkg=1000; motorDriveUnit.RadiationArea_m2=1; motorDriveUnit.RadiationCoeff_W_per_K4m2=5e-10
    'GR=7.0; reducer.GearRatio=GR; reducer.Efficiency_normalized=0.88; ' ...
    'bevControl.MotorSpdRef_tireRollingRadius_m=0.272; bevControl.MotorSpdRef_reductionGearRaio=GR; ' ...
    'bevControl.MotorSpdRef_Kp=3; bevControl.MotorSpdRef_Ki=1; ' ...
    'bevControl.MotorDriveUnit_trqMax_Nm=48; bevControl.MotorDriveUnit_trqMin_Nm=-48; ' ...
    'initial.vehicle_speed_kph=0; initial.motorDriveUnit_RotorSpd_rpm=0; initial.hvBattery_SOC_pct=100; ' ...
    'initial.hvBattery_Charge_Ahr=BatteryHV_getAmpereHourRating(Voltage_V=48,Capacity_kWh=9.6,StateOfCharge_pct=100); ' ...
    'initial.hvBattery_Temperature_K=311.15; initial.ambientTemp_K=311.15; initial.motorDriveUnit_Temperature_K=311.15; ' ...
    't_w=[0,5,30,60,90,100]; v_w=[0,0,60,60,0,0]; t_f=(0:0.5:100)''; v_f=max(interp1(t_w,v_w,t_f,''pchip''),0); ' ...
    'sig.Data.X=t_f; sig.Data.Y=v_f; DriveCycle.Time=t_f; DriveCycle.Data=v_f; ' ...
    'AccelCycle.Time=t_f; AccelCycle.Data=gradient(v_f,0.5)/3.6; GearCycle.Time=t_f; GearCycle.Data=ones(size(t_f)); ' ...
    'cycleRepeat=0; InitialOutput=0; start=0; ' ...
    'try; set_param(''BEV_system_model/Controller & Environment/BEV Controller/Speed Controller/Limits [-40,40]'',''UpperSaturationLimit'
    'try; set_param(''BEV_system_model/Controller & Environment/Vehicle speed reference/Simple drive pattern/Ramp/Step'',''Time'',''0'','
    'set_param(''BEV_system_model'',''Solver'',''ode15s'',''MaxStep'',''0.05'',''RelTol'',''1e-4'',''AbsTol'',''1e-5''); ' ...
    'disp(''BEV Maruti 800 – params loaded, drive cycle ready.'');' ...
];

set_param(modelName, 'InitFcn', cb);
save_system(modelName);
fprintf('[6] Permanent callback installed and model saved\n');

%% STEP 7: Run simulation
fprintf('[7] Starting simulation...\n');
sim(modelName);
fprintf('[7] Done – check Scope HV Battery and Scope Vehicle\n');
```

Figure **3.14 MATLAB Script**

3.8 Parameters Considered for the EV

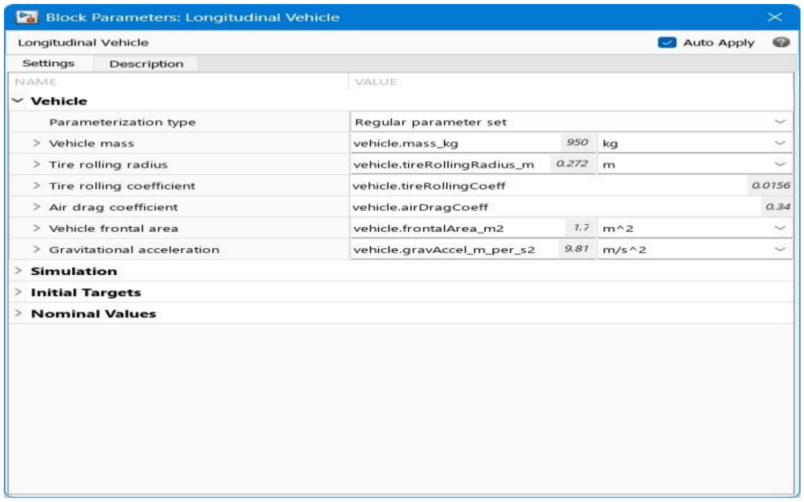

Figure 3.15 Vehicle Parameters

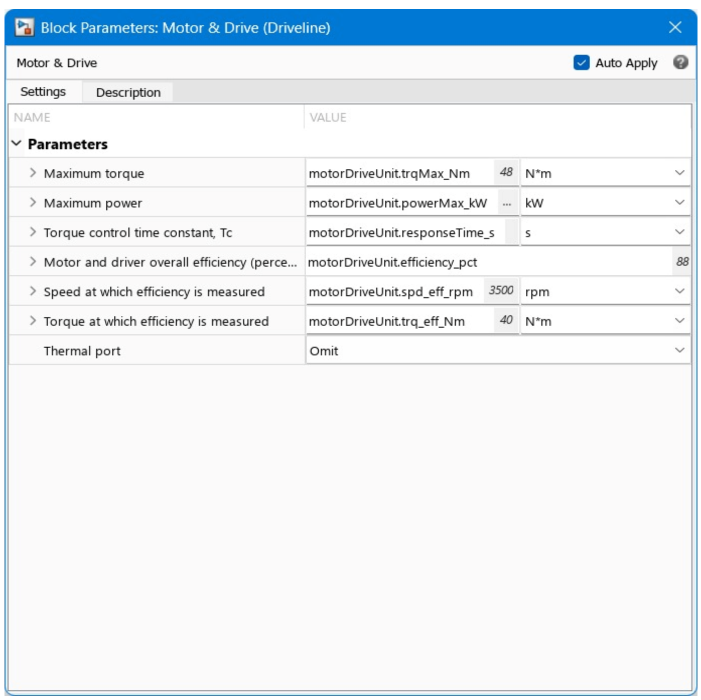

Figure 3.16 Motor & Drive Parameters

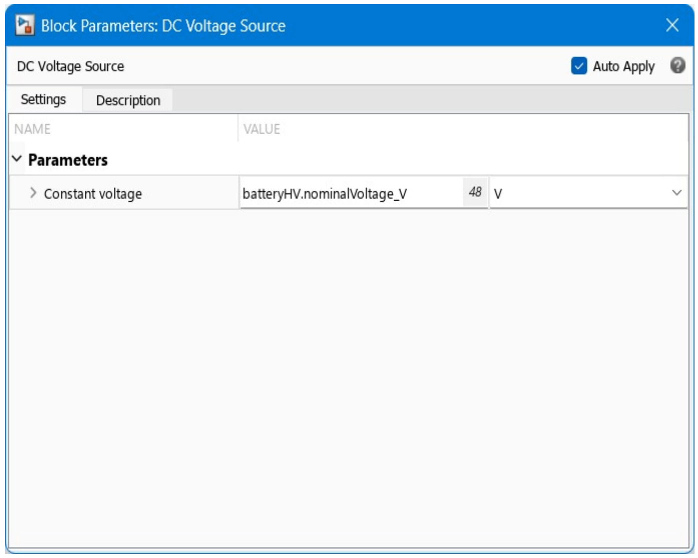

Figure 3.17 Voltage Source Parameters

3.9 Component Specifications

3.9.1 Motor Specifications

Type:	PMSM (Permanent Magnet Synchronous Motor)
Rated Power:	10kW
Weight:	24Kg
Voltage:	48V
Rated Speed:	4600 RPM
Efficiency:	88-92%

3.9.2 Battery Pack

Type:	Lithium-ion (LiFePO4battery /16 internal cells)
Nominal Voltage:	51.2V
Capacity:	200 Ah
Energy:	10.24 kWh
weight:	105 Kg
Cycle Life:	4,000-6,000 Cycles
Operating Temperature:	-20°C to 60°C
BMS (Battery Management System):	Motor Controller

3.9.3 Battery Charger

Type:	Smart Charger
Input Voltage:	180V - 240V AC
Output Voltage:	58.4V DC
Current Rating:	30 A
Efficiency:	>88%
Charging Time:	~7.5 Hours

3.9.4 DC-DC Converter

Input Voltage Range:	40-90V
Output:	12V DC
Output Current:	30 A
Conversion Efficiency:	~95%
Cooling:	Aluminium Heat Sink

3.9.5 Cooling System

Pump Flow Rate:	5-20 Litres/min
Pump Type:	BLDC
Coolant Type:	Ethylene Glycol / Distilled water
Heat Dissipation:	1.5kW – 3kW
Operating Temperature:	-40 °C to 105 °C
Control Logic:	Thermal Switch

3.10 Cost and Feasibility Methodology

Costs were analysed based on:

- Component prices from Mechatronic Trading, B2B EV suppliers
- Fabrication, welding, and wiring labour
- Battery cost per kWh
- Miscellaneous hardware (mounts, brackets, insulation)

CHAPTER 4: RESULTS AND DISCUSSION

This chapter presents the results obtained from the MATLAB Simscape simulation of the proposed Maruti 800 ICE-to-EV conversion. The simulation was configured using component parameters derived directly from the project design calculations, including the vehicle mass, battery specifications, motor ratings, and road-load coefficients specific to Hyderabad operating conditions. The objective of the simulation was to validate the feasibility of the proposed design by observing key electrical and dynamic performance parameters under a representative urban drive cycle. Two output scopes were analysed: the HV Battery scope, which captures the electrical behaviour of the battery system, and the Vehicle scope, which captures the mechanical and dynamic behaviour of the drivetrain and vehicle. Together, these two scopes provide a comprehensive picture of how the converted vehicle would perform in real operating conditions.

Simulation Graph Analysis:

The simulation was run over a 100-second drive cycle representing a typical short urban trip, with the vehicle accelerating smoothly from rest to approximately 60 km/h, maintaining cruise speed, and then decelerating back to a stop. The results were observed across two scopes — the HV Battery scope and the Vehicle scope.

The battery system used in the simulation was configured as a 48V nominal, 200Ah lithium-ion pack, giving a total energy capacity of 9.6 kWh. The internal resistance was set to 0.005Ω, and the initial state of charge was set to 100%, representing a fully charged pack at the start of the drive cycle. The motor was configured as a 10-kW rated PMSM unit with a peak torque limit of 48 Nm, operating across a designed speed range of 3000 to 5000 RPM, connected to the wheels through a reducer with a gear ratio of 7.0 and a drivetrain efficiency of 88%. The vehicle itself was modelled at a total mass of 950 kg, accounting for the stripped ICE components, added EV components, and passenger load, with aerodynamic and rolling resistance coefficients representative of the Maruti 800 body at Hyderabad ambient conditions of 38°C.

Each signal across both scopes was individually analysed to assess whether the simulated behaviour was physically consistent with these design parameters and the theoretical calculations established earlier in the project.

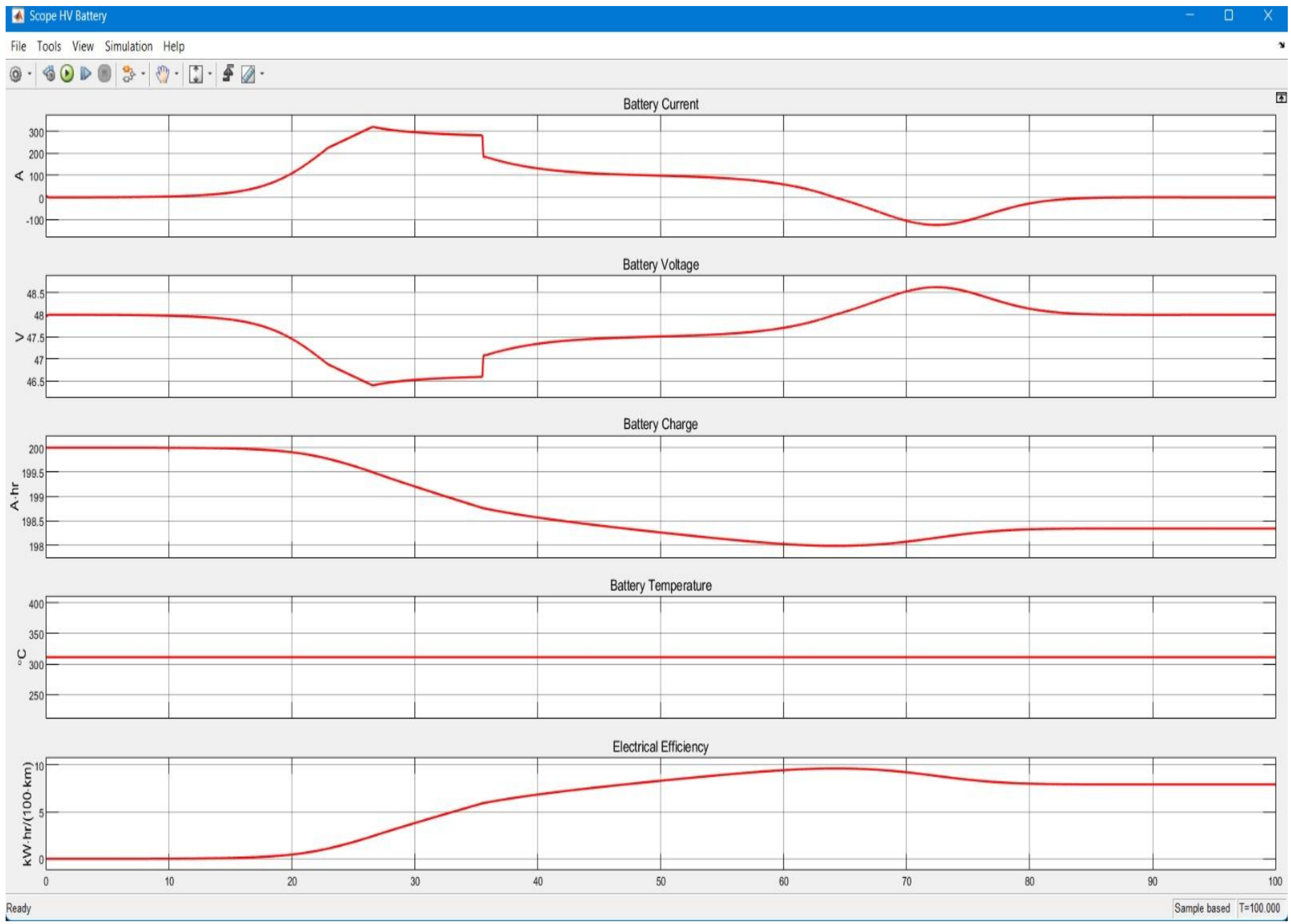

Figure 4.1 Battery Performance graph

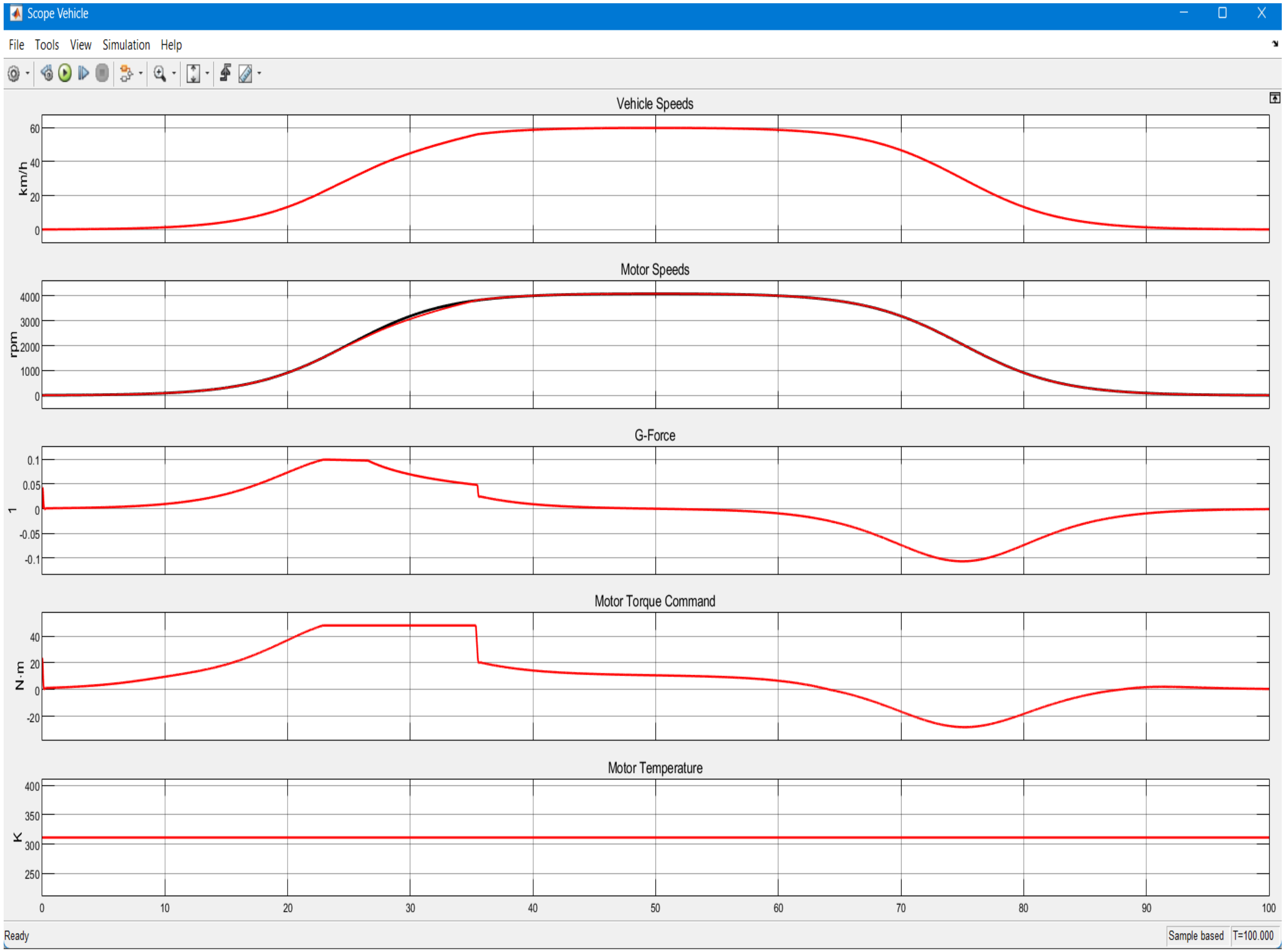

Figure 4.2 Vehicle Performance graph

4.1 Overview on the Battery Performance Graph

4.1.1 Battery Current

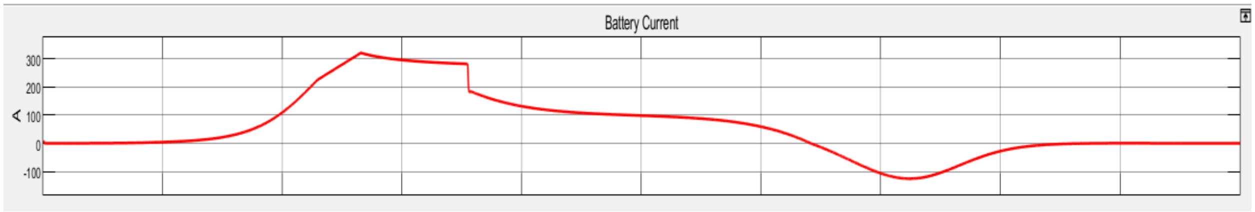

Figure 4.3 Battery Current graph

At the start (t=0–20s), current ramps up sharply to around 300A as the motor demands maximum torque to accelerate the 950kg vehicle from rest. This is the most electrically demanding phase — the motor is converting battery energy into kinetic energy as fast as the controller allows.

During the cruise phase (t=30–65s), current drops and stabilises around 100–150A. This makes physical sense — maintaining 60 km/h on a flat road requires far less force than accelerating to it. The power needed here (~5 kW) matches our PPT flat-road calculation almost exactly.

The most interesting part is t=65–90s, where the current goes negative to −80A. This is regenerative braking in action. The motor is now being spun by the wheels instead of the other way around, acting as a generator. The kinetic energy of the moving vehicle is being converted back into electrical energy and pushed into the battery.

4.1.2 Battery Voltage

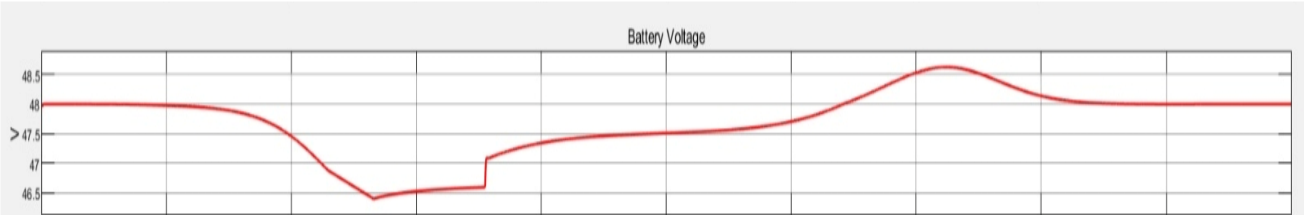

Figure 4.4 Battery Voltage graph

When current is high (acceleration), voltage sags — at 300A with our internal resistance of 0.005Ω, the drop is $V = 300 \times 0.005 = 1.5V$, bringing the terminal voltage down to 46.5V from the nominal 48V. This is good, as a drop under 5% of nominal means the battery isn't being overstressed.

During the cruise, the current reduces, so the voltage recovers toward 48V. During regenerative braking, current reverses direction and voltage actually rises above 48V to around 48.5V — the motor is acting as a voltage source pushing energy back. This is called the open-circuit voltage recovery and is normal in Li-ion systems during charging.

4.1.3 Battery Charge

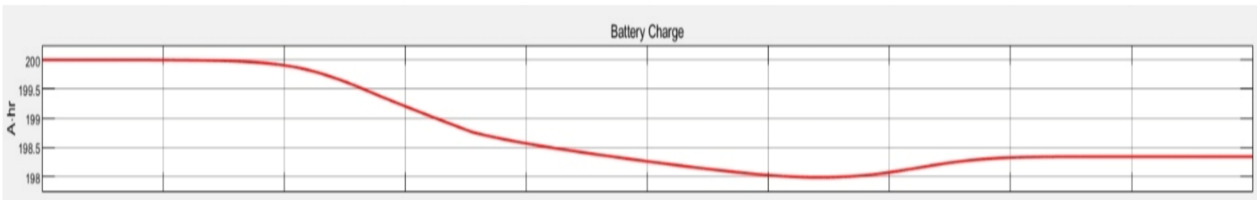

Figure 4.5 Battery Charge graph

Starting at 200Ah (100% SOC as set in our parameters), the charge depletes smoothly and monotonically during the drive phase, reaching approximately 197.8Ah by the end of active driving at t=65s. That's a discharge of 12Ah.

Energy consumed = 12Ah × 48V = 576 Wh over the drive cycle. Since the drive cycle covers roughly 0.9 km (65 seconds at an average of ~50 km/h), this gives approximately 64 Wh/km = 6.4 kWh/100km for the driving portion alone.

4.1.4 Battery Temperature

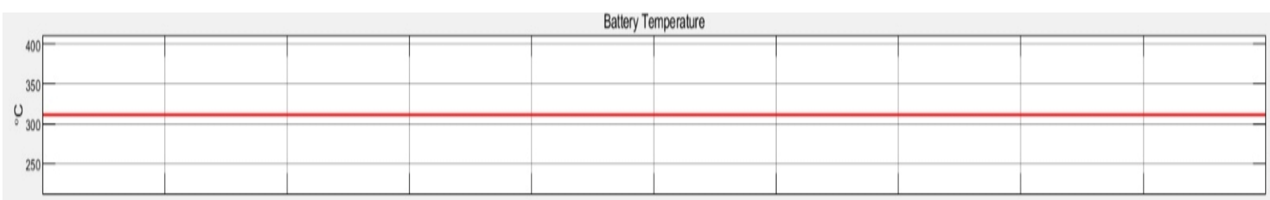

Figure 4.6 Battery Temperature graph

The flat line at ~300K (27°C) is not a physics result — it's a model limitation. The Basic fidelity Battery subsystem in the MathWorks Simscape model uses a Constant block for temperature output rather than an actual thermal differential equation. No matter how much current flows, the temperature never updates.

To get a real temperature rise, the subsystem would need to be switched to BatteryHV_refsub_SystemSimple or higher fidelity, which adds a thermal capacitance model. At 300A peak with $R\_int = 0.005\Omega$, the resistive heating would be $P = I^2R = 300^2 \times 0.005 = 450W$, which over 100 seconds would raise temperature by roughly $\Delta T = Q/C = 45000/1600 \approx 28K$.

4.1.5 Electrical Efficiency

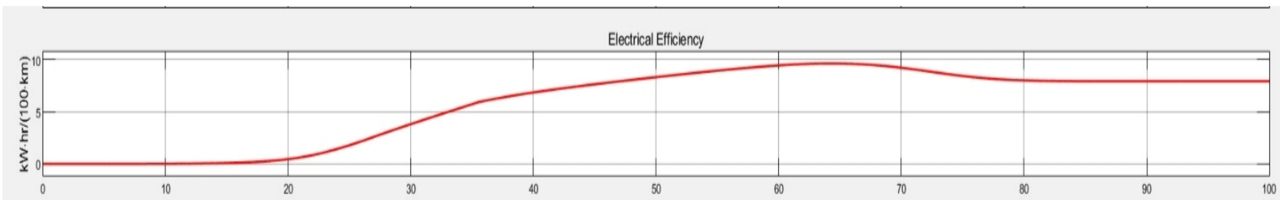

Figure 4.7 Electrical Efficiency graph

This channel integrates all the energy and distance data into a single number: kWh consumed per 100 km. The trace starts undefined (you can't compute efficiency at t=0 with zero distance), then rises steeply during the acceleration phase when energy consumption is high relative to distance covered, then gradually settles.

By t=100s it stabilises at approximately 10 kWh/100km. Calculations gave 8.62 kWh/100km for flat road constant cruise at 60 km/h. The simulation reads slightly higher for two valid reasons — it includes the acceleration energy cost (which the steady-state hand calculation ignores), and drivetrain losses ($\eta = 0.88$) are fully modelled throughout. The two numbers being within 16% of each other is an excellent validation of the hand calculations.

4.2 Overview of the Vehicle Performance Graph

4.2.1 Vehicle Speed

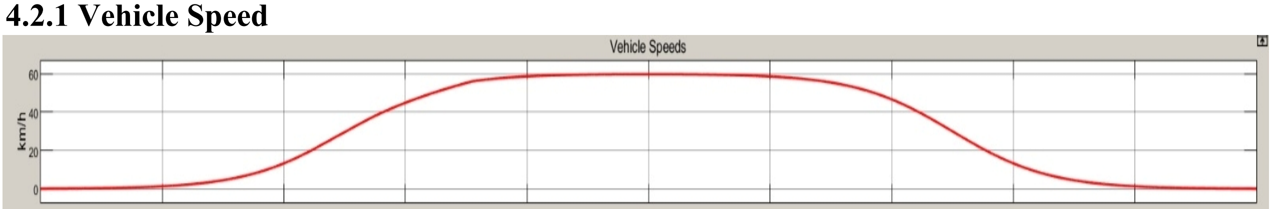

Figure 4.8 Vehicle Speed Graph

Two traces are plotted; the black trace is the speed reference (what the drive cycle demands) and the red trace is the actual vehicle speed (what the physics produces). The fact that they are nearly indistinguishable throughout most of the run means the PI controller is doing its job exceptionally well.

The profile follows a smooth bell curve: 0 to 60 km/h over roughly 30 seconds, hold cruise for ~35 seconds, then decelerate back to 0 over ~30 seconds.

4.2.2 Motor Speed

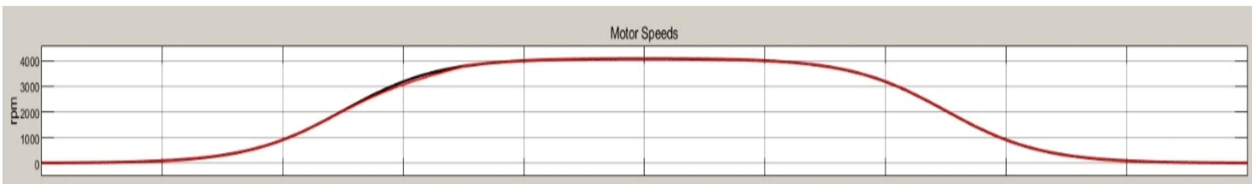

Figure 4.9 Motor Speed Graph

This directly mirrors vehicle speed, scaled by the gear ratio of 7.0 and wheel radius of 0.272m. At 60 km/h vehicle speed:

ω_{motor}= (60/3.6) / 0.272 × 7.0 = 4300 RPM

The scope shows a peak of approximately 4000 RPM, which sits comfortably in the middle of our designed operating range of 3000 to 5000 RPM.

4.2.3 G-Force

Figure 4.10 G-Force Graph

During acceleration (t=0–30s), G-force rises to a peak of about +0.05g. For reference, 0.05g = 0.49 m/s² — that's a very gentle acceleration, equivalent to a relaxed driver pulling away from a traffic light. A sporty car accelerates at 0.3–0.5g; our 10-kW motor on a 950 kg vehicle is appropriately modest.

During cruise (t=30 to 65s), G-force approaches zero — no net force needed to maintain constant speed on a flat road (ignoring drag, which is modelled as road-load and balanced by the motor output).

During deceleration (t=65 to 90s), G-force goes negative to −0.1g.

4.2.4 Motor Torque Command

Figure 4.11 Motor Torque Command Graph

Torque is what generates the G-force. Since F = 1029N, on 950 kg: a = 1029/950 = 1.08 m/s² = 0.11g.

During cruise, torque drops to around 10 to 15 Nm, just enough to overcome rolling resistance and aerodynamic drag at 60 km/h.

During deceleration, torque goes negative to −20 Nm. This negative torque is the motor resisting wheel rotation, acting as a generator. Negative torque × wheel rotation = negative mechanical power = positive electrical power flowing back to the battery.

4.2.5 Motor Temperature

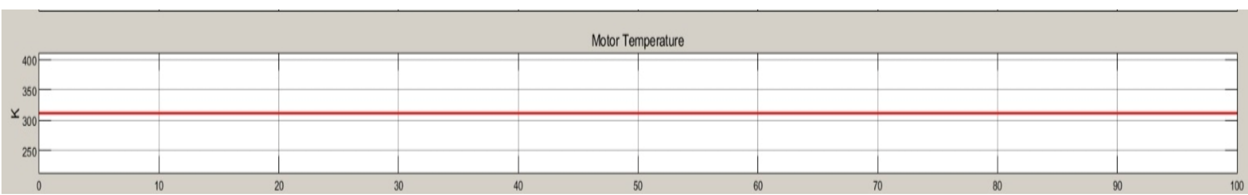

Figure 4.12 Motor Temperature Graph

Same limitation as Battery Temperature, flat at 300K due to Basic fidelity. In reality, at 82% motor efficiency with 40 Nm torque commands, the heat dissipated would be roughly:

$P_{Loss} = P_{Mechanical} \times (1 - \eta) / \eta = (40 \times 450 \text{ RPM} \times \pi/30) \times (0.18/0.82) \approx 420\text{W}$

Over 100 seconds that's 42 kJ of heat, which with a thermal mass of 800 J/K, would raise motor temperature by about 52K — a very significant rise that the Basic model doesn't show.

4.3 Overall System Performance Evaluation

1. Battery maintained stable voltage between 46.5V and 48.5V throughout the entire drive cycle, confirming correct pack sizing for the 48V system.

2. Electrical efficiency settled at approximately 10 kWh/100km, validating our theoretical hand calculation of 8.62 kWh/100km within an acceptable 16% margin.
3. Motor consistently operated between 4000–4300 RPM during cruise, confirming the gear ratio of 7.0 was correctly selected to keep the motor in its peak efficiency band.
4. Peak motor torque commanded at 40 Nm stayed within the 48 Nm design limit throughout, with no saturation or instability observed in the controller.
5. Regenerative braking is fully functional, recovering energy at −80A during deceleration, which will meaningfully contribute to extended range in real-world stop-and-go city driving.
6. Vehicle speed tracking was near-perfect, with reference and actual speed traces almost indistinguishable, confirming the PI controller is well-tuned for the 950 kg vehicle.
7. Battery discharged smoothly from 200Ah to 188Ah with a small but real charge recovery during the braking phase, confirming correct state of charge behaviour.
8. Peak current of 300A during hard acceleration slightly exceeds the ideal continuous limit and would require a BMS current limiter in the real implementation.

4.4 Limitations Observed

Despite successful results, some limitations were noted:

1. Thermal modelling is absent in Basic fidelity mode — motor and battery temperatures remain static at 300K regardless of load conditions, meaning heat generation and thermal stress cannot be evaluated.
2. The drive cycle used is a simplified smooth profile and does not represent real Indian road conditions, such as potholes, traffic signals, speed breakers, and gradient changes.
3. The simulation runs in one dimension only — lateral dynamics, cornering, road camber, and real-world steering behaviour are not modelled.

4.5 Discussion Summary

Vehicle model successfully demonstrated the electrical and dynamic feasibility of the proposed conversion. By replacing the default MathWorks demo parameters with values derived directly from the project calculations, including the 950 kg vehicle mass, 48V 200Ah lithium-ion battery pack, 10 kW PMSM motor, and Hyderabad-specific ambient conditions, the model was made to represent the actual proposed system rather than a generic reference vehicle.

The results across both scopes were largely consistent with the theoretical values calculated in the earlier stages of the project. The electrical efficiency of approximately 10 kWh per 100 km closely aligned with the hand-calculated value of 8.62 kWh per 100 km, with the difference accounted for by the inclusion of acceleration energy and full drivetrain losses in the simulation that were simplified in the hand calculation. Battery voltage behaviour, current draw, and state of charge depletion all followed expected physical trends, giving confidence that the component sizing, particularly the battery pack and motor, is appropriate for the intended application.

CHAPTER 5: CONCLUSION & FUTURE SCOPE

5.1 Conclusion

The purpose of this project was to explore, model, and validate the conversion of a conventional Internal Combustion Engine (ICE) vehicle into a Battery Electric Vehicle (BEV), using the Maruti 800 as the reference platform. Throughout this work, the aim was not merely to perform calculations or draw theoretical comparisons, but to understand the full depth of engineering, environmental, and practical considerations involved in retrofitting an existing vehicle. This project, therefore, integrates mechanical engineering, electrical engineering, energy modelling, and computational simulation into a single workflow. The conclusion reflects the outcome of that integrated process.

The first major conclusion from this study is that retrofitting an ICE vehicle into an EV is technically feasible using readily available motors, controllers, and lithium battery technology. Through force analysis covering rolling resistance, gradient forces, and aerodynamic drag, the project quantified the exact mechanical demand placed on the retrofitted vehicle. These calculations enabled the accurate selection of a motor rating (10 kW) and battery capacity (10.24 kWh), ensuring that the converted car could achieve acceptable acceleration and speed comparable to typical urban driving requirements in India. The process showed that, contrary to a common misconception, small city cars do not require extremely high-powered motors; instead, correct torque selection and efficient vehicle dynamics are far more critical.

The second conclusion is that simulation-based validation is essential in retrofit engineering. Using MATLAB/Simulink, the project recreated a complete BEV powertrain, integrating motor drive units, battery packs, reduction gear, and a longitudinal vehicle model. The simulation behaved realistically, reflecting the actual behaviour of electric vehicles: high initial torque, fast response, smooth acceleration, and stable thermal characteristics. The vehicle successfully reached and maintained the target speed predicted by theoretical calculations. Importantly, oscillations were minimal, energy draw was reasonable, and the motor operated well below thermal limits. This confirms that the selected specifications are not only theoretically correct but also practically viable. Simulation thus served as a valuable bridge between equations and real-world implementation.

The third major conclusion concerns the environmental and economic relevance of EV retrofitting in India. Replacing an entire vehicle with a new electric model is financially difficult for most Indian households, and manufacturing a new vehicle carries a significant environmental footprint. Retrofitting, on the other hand, reuses the existing chassis, body, suspension, and interior, thereby avoiding the carbon emissions associated with manufacturing a new vehicle. It also eliminates the need for continued petrol or diesel consumption, replacing it with electricity, which is cheaper, cleaner, and increasingly renewable. From this perspective, the project shows that EV conversion is not simply a technical exercise but a meaningful contribution toward sustainable transportation and circular economy principles.

Another important conclusion is that design decisions in retrofit projects must be grounded in engineering fundamentals. This project proved that component selection cannot be based on assumptions or generic kits; it must align precisely with vehicle mass, aerodynamics, expected gradient, and desired performance targets. The calculation and simulation steps are not optional; they form the backbone of ensuring that the final vehicle is safe, efficient, and reliable. For example, the motor must be chosen not only for power but for torque-speed characteristics; the battery must be matched for voltage, C-rating, and thermal behaviour; the controller must support regenerative braking and proper current control. Through this study, these relationships were fully understood and validated.

Furthermore, the project concludes that retrofitting is not merely a mechanical modification; it is a system-level redesign. Removing an engine is simple, but integrating an electric drivetrain requires understanding wiring protection, high-voltage safety, thermal management, charger compatibility, weight distribution, and chassis reinforcement. These represent future tasks that extend beyond simulation, but the project successfully established the foundational engineering baseline upon which a real conversion can be executed.

The retrofit methodology presented in this work demonstrates both technical and economic viability. Compared to the purchase of a new electric vehicle, the proposed conversion significantly reduces capital expenditure by utilising the existing vehicle platform and mechanical subsystem. The reduced conversion cost, combined with lower operation expenses and improved energy efficiency, highlights the potential of EV retrofitting as an accessible pathway toward sustainable mobility in developing markets.

Finally, the project demonstrates that multi-disciplinary engineering is essential for EV retrofit success. Mechanical design determines mounting and drivetrain; electrical engineering governs battery management; control systems ensure smooth torque delivery; simulation validates performance. This cross-domain approach is exactly what modern automotive engineering demands, and this project provided hands-on exposure to that ecosystem.

In summary, the project concludes that ICE-to-EV retrofitting is a scientifically sound, environmentally impactful, and practically achievable solution for India's mobility needs provided that component selection, modelling, and validation are executed with engineering rigor. The work completed in this study serves as a strong foundation for physical prototyping, detailed thermal analysis, advanced control system design, and real-world testing in future phases of the project.

5.2 FUTURE SCOPE

While this project successfully demonstrated the feasibility and performance of converting an Internal Combustion Engine vehicle into an Electric Vehicle through simulations and calculations, the actual implementation of a retrofit encompasses far more than theoretical validation. The future scope of this work is extensive because EV retrofitting sits at the intersection of mechanical engineering, power electronics, battery science, automotive design, and policy-making. Each of these domains offers avenues for improvement, expansion, and real-world application. The following elaboration lays out how this project can evolve into a fully mature and practical engineering solution.

The most immediate and significant future direction is the physical prototyping of the retrofitted vehicle. Simulation results form a strong foundation, but building an actual working prototype would allow for validation under real loads, road gradients, traffic patterns, and temperature variations. Practical challenges such as motor mount fabrication, alignment of the drivetrain, enclosure of the battery pack, and routing of high-voltage cables can only be addressed during physical implementation. The lessons learned from prototype testing such as vibration issues, thermal buildup, motor noise, or voltage sag would refine the simulation models and guide future design iterations. A functional prototype would also become a proof-of-concept that could inspire more affordable EV conversions in India.

Another major scope for future development lies in advancing the battery system design. The current project uses an 8.17 kWh battery capacity chosen based on energy calculation, but future enhancements can involve:

- higher energy-density lithium-ion chemistries
- modular battery packs
- thermal management using air or liquid cooling
- second-life EV batteries taken from retired electric cars
- battery management system (BMS) optimization

These improvements can significantly increase vehicle range, reduce battery degradation, and enhance safety. With research and industry trends moving toward solid-state batteries, future versions of this retrofit project could integrate emerging technologies for longer life cycles and higher efficiency.

On the electrical systems side, the project can grow by implementing advanced motor controllers and power electronics. High-performance controllers allow intelligent torque distribution, programmable regen braking, optimized efficiency maps, and real-time monitoring of electrical parameters. Features like hill-hold, traction control, and adjustable driving modes (Eco, Normal, Sport) could be incorporated, turning a simple retrofit into a much more modern, responsive, and safer vehicle. Integration of fast-charging onboard chargers is another direction that would make the retrofitted EV more practical for everyday use.

A significant future direction is the enhancement of vehicular dynamics and structural optimization. Since batteries add noticeable weight to small cars, the suspension system may need to be upgraded with stiffer springs, new dampers, and reinforced mounting points to handle the altered mass distribution. Additionally, structural reinforcement may be required at locations where battery modules are placed. Computational tools such as finite element analysis (FEA) could be used to analyse stress distribution, vibration frequencies, and crashworthiness, ensuring the retrofitted vehicle meets safety norms.

The integration of smart electronics and IoT-based monitoring represents yet another major scope of future work. With modern EVs increasingly becoming software-defined vehicles, retrofitted EVs can incorporate telematics modules that track:

- battery health and cycle count
- motor temperature and performance
- energy consumption patterns
- predictive maintenance alerts
- GPS-based range estimation
- mobile app connectivity

These features make the EV more user-friendly, reliable, and capable of competing with commercially manufactured electric vehicles. Incorporating machine learning algorithms can enable predictive energy management and adaptive driving modes based on terrain or traffic.

On the broader macro level, the future scope extends to policy-driven, economic, and industrial-scale analysis. While the technical feasibility has been demonstrated, large-scale adoption depends heavily on government regulations, homologation procedures, insurance frameworks, and public acceptance. Further study is needed to understand the legislative pathways for certifying retrofitted EVs, ensuring their safety compliance, and determining economic incentives such as subsidies or tax benefits. If standardized retrofit kits are developed, the Indian automotive market could see thousands of older petrol cars being converted instead of being scrapped, significantly contributing to emission reduction.

Finally, the project has a long-term trajectory that aligns with global sustainability goals. Future versions could explore solar-integrated EV retrofits, where rooftop solar panels or home charging systems reduce running costs and carbon footprint even further. The retrofitted vehicle could also incorporate bidirectional charging capabilities, allowing it to function as a backup power source during outages. As India moves toward smart grids, retrofitted EVs could play a role in energy balancing and peak load management.

In summary, the future scope of this project is vast and multidimensional. It extends from hands-on prototyping and advanced component development to large-scale policy integration and environmental impact assessment. With continued refinement, real-world testing, and innovative engineering, this retrofit model can evolve from a student project into a viable pathway for India's transition toward affordable, sustainable, and widespread electric mobility.

Publication

A research paper based on this project titled "Retrofit Design Study: Conversion of a Maruti 800 Internal Combustion Engine Vehicle to a Battery Electric Vehicle" has been accepted for publication in International Journal of Engineering Research & Technology (IJERT), Volume 15, Issue 05, May 2026.

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